

ASTR 302 Fall 2026

Uncertainties and Statistics Introduction



Professor Brittany E. Miles
Steward Observatory

Agenda – August 26, 2026

- Why Measurement Error and Statistics Matters
- How do Astronomers Estimate and Report Errors
- Sources of Error in Astronomical Observations
- Probability Distributions

FRIDAY BRING A LAPTOP OR COMPUTER

Recommended Reading: Taylor Chapter 1 and 2

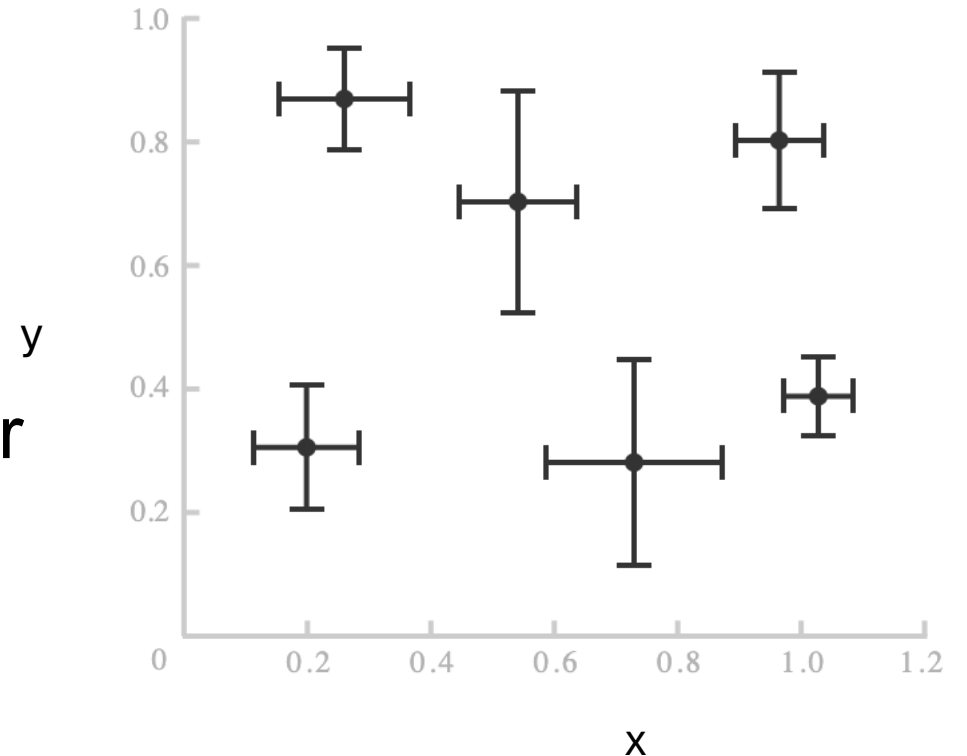
What is Measurement Uncertainty

Uncertainty (σ) describes the probable range of values associated with a measurement. Measurement uncertainty is often referred to as “**error**” or “**noise**” in scientific papers.

$$26 M_{jup} \pm 5$$

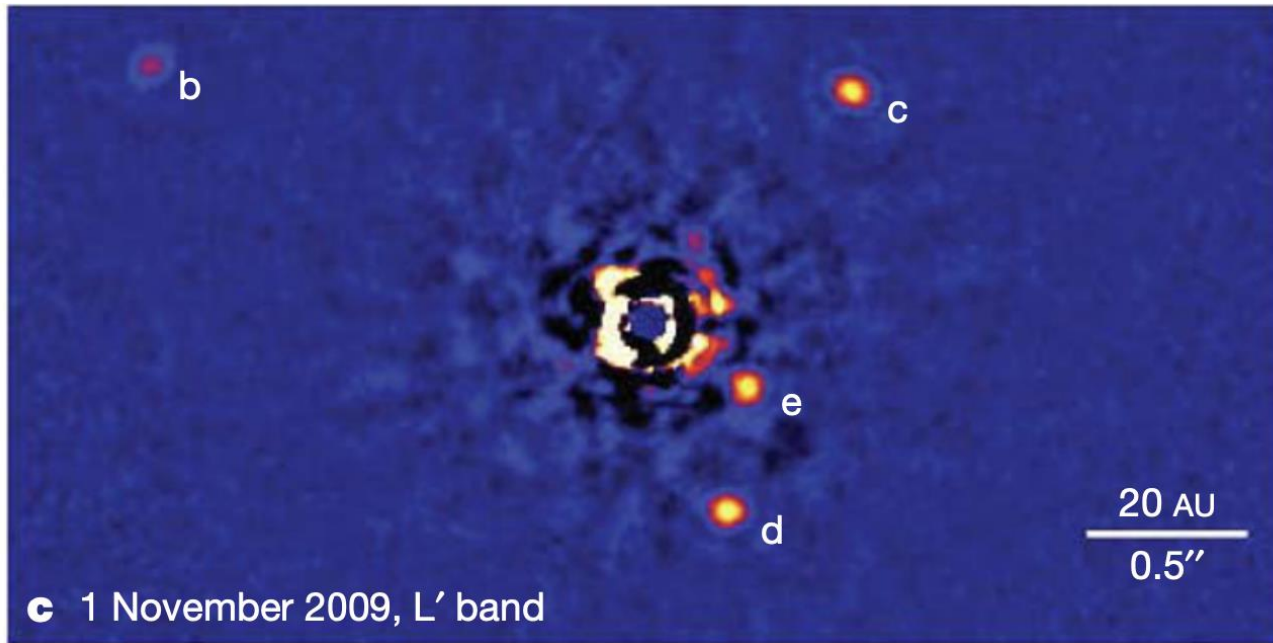
$$52 M_{\odot} {}^{+5}_{-2}$$

Signal-to-noise (SNR) is the ratio of your measurement /signal (μ) to the noise (σ).



What is Measurement Uncertainty

In astronomy we often report derived quantities. The total noise of a quantity is often the combination of multiple effects. We must understand and account from them carefully.



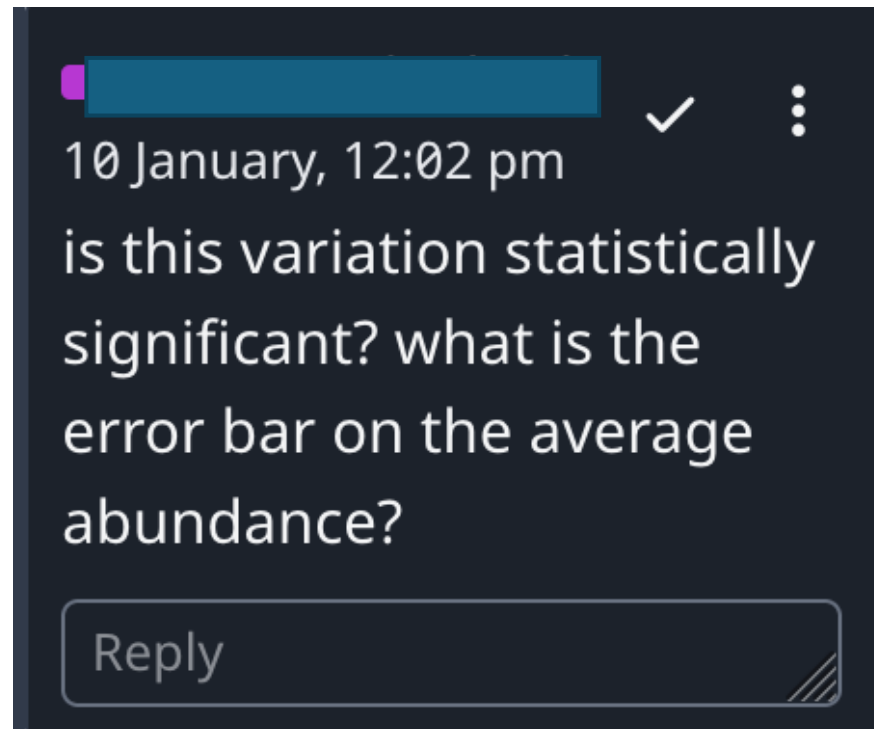
HR 8799 planetary system, Marois+ 2010



Stephan's Quintet seen
from JWST/NIRCam

Why Measurement Error and Statistics Matters

- You need to know how confidently something was measured.
-  I found oxygen on an exoplanet.
-  I measured an oxygen abundance of 500ppm +/- 200ppm on an exoplanet



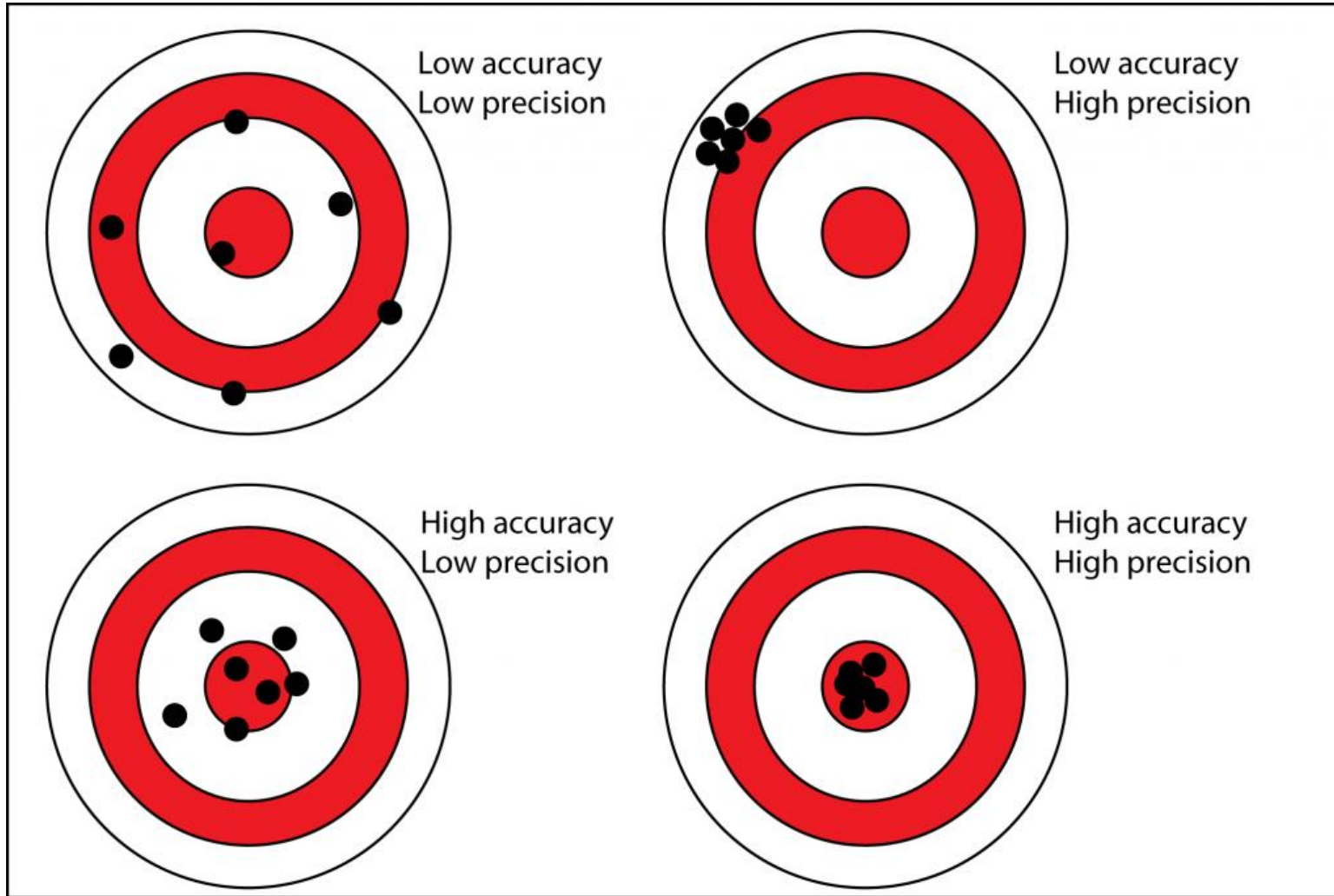
Why Measurement Error and Statistics Matters

- Measurement error determines how well you can compare your data to theoretical models and test a specific hypothesis.
- Understanding what causes the most noise/error in your experiment is important for improving future tests or observations.
- You can use these skills to be a critical reader and informed citizen.

How Do Astronomers Estimate Errors

- **Error propagation** combines the individual errors from each variable that contributes to a derived quantity. This requires understanding where errors are introduced in your telescope and instrument.
- **Statistics** rely upon performing the experiment numerous times, where the value for the derived variable is the mean of the individual estimated values and the uncertainty is the standard deviation about that mean.

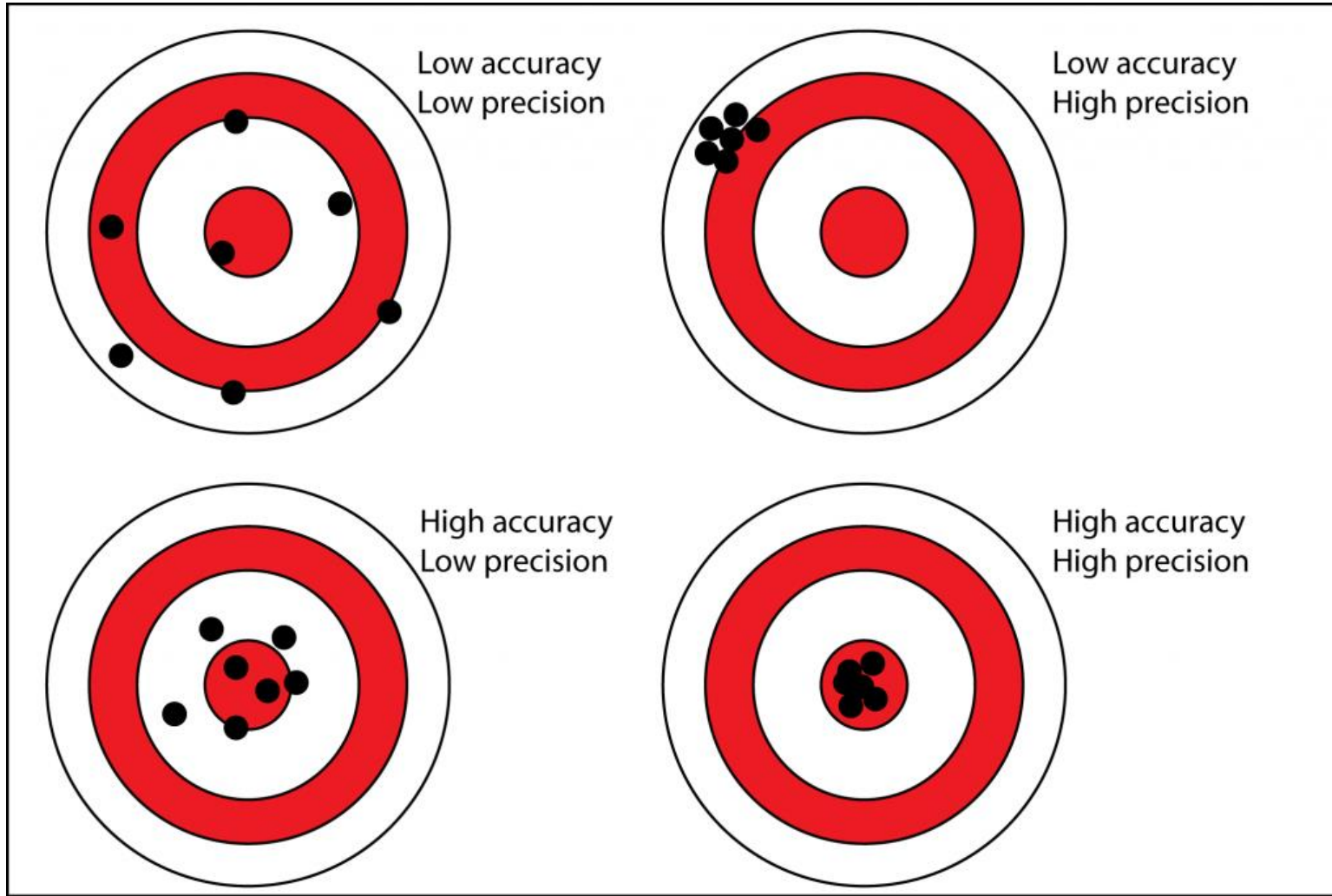
Precision vs Accuracy



Precision is a measure of repeatability of a measurement.

Accuracy is a measure of how well the data matches the expected value.

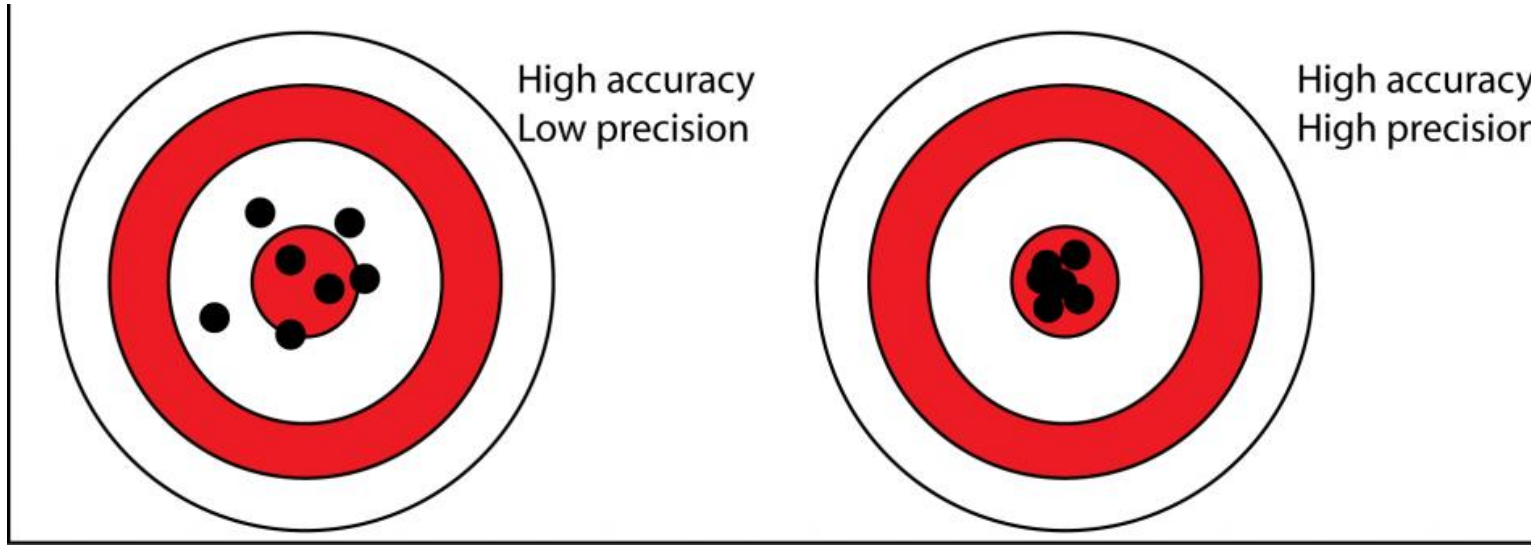
Precision vs Accuracy in Astronomy



Precision is often determined by the astronomical source, telescope, atmosphere, and calibration method.

Accuracy is often determined by systematic errors from a telescope, atmosphere, or calibration method.

Estimating Expected Values and Error

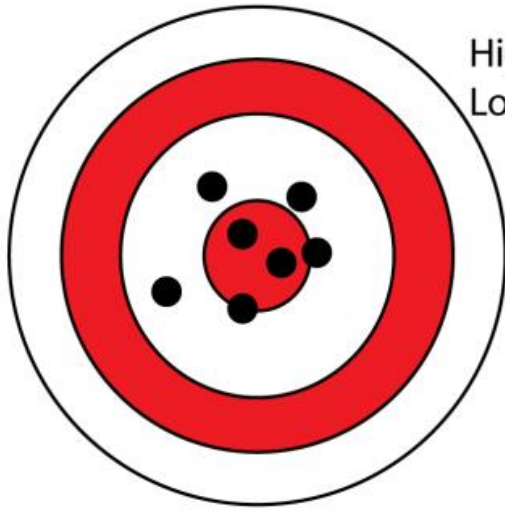


The expected value can be estimated by taking the mean of a set of data points.

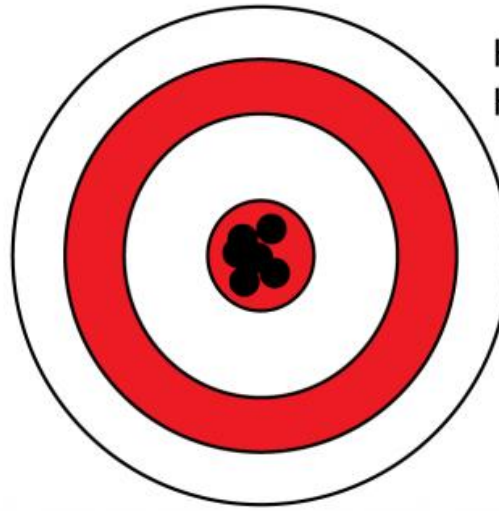
$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i = \frac{x_1 + x_2 + \cdots + x_n}{n}$$

See `numpy.mean` (python)

Estimating Expected Values and Error



High accuracy
Low precision



High accuracy
High precision

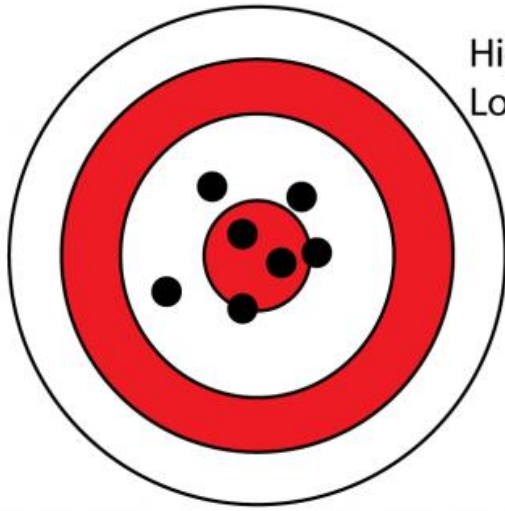
The expected value can be estimated by taking the mean of a set of data points.



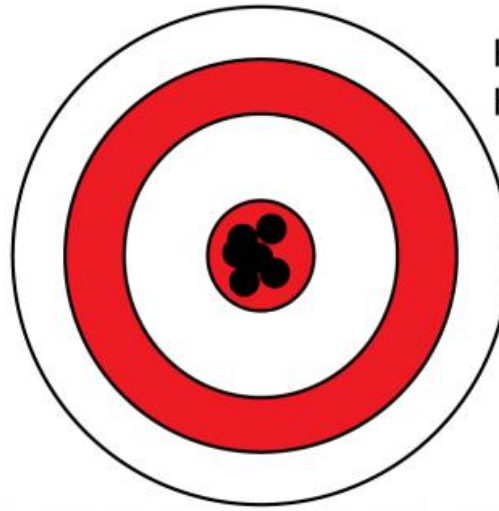
$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i = \frac{x_1 + x_2 + \cdots + x_n}{n}$$

See `numpy.mean` (python)

Estimating Expected Values and Error

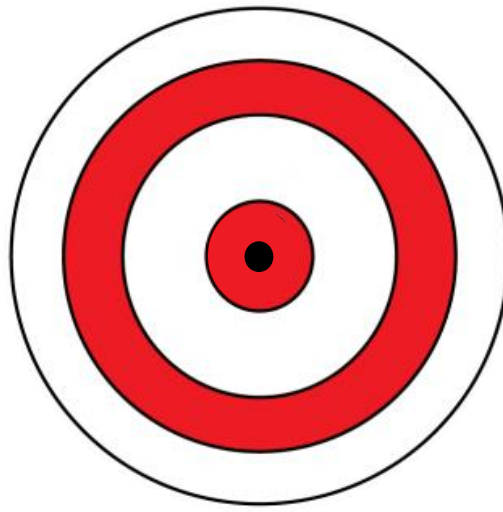
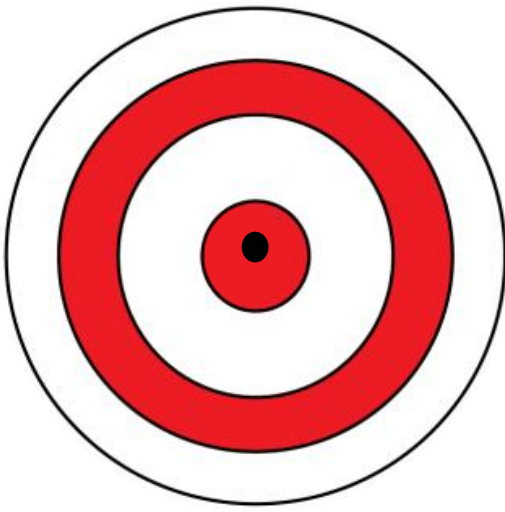


High accuracy
Low precision



High accuracy
High precision

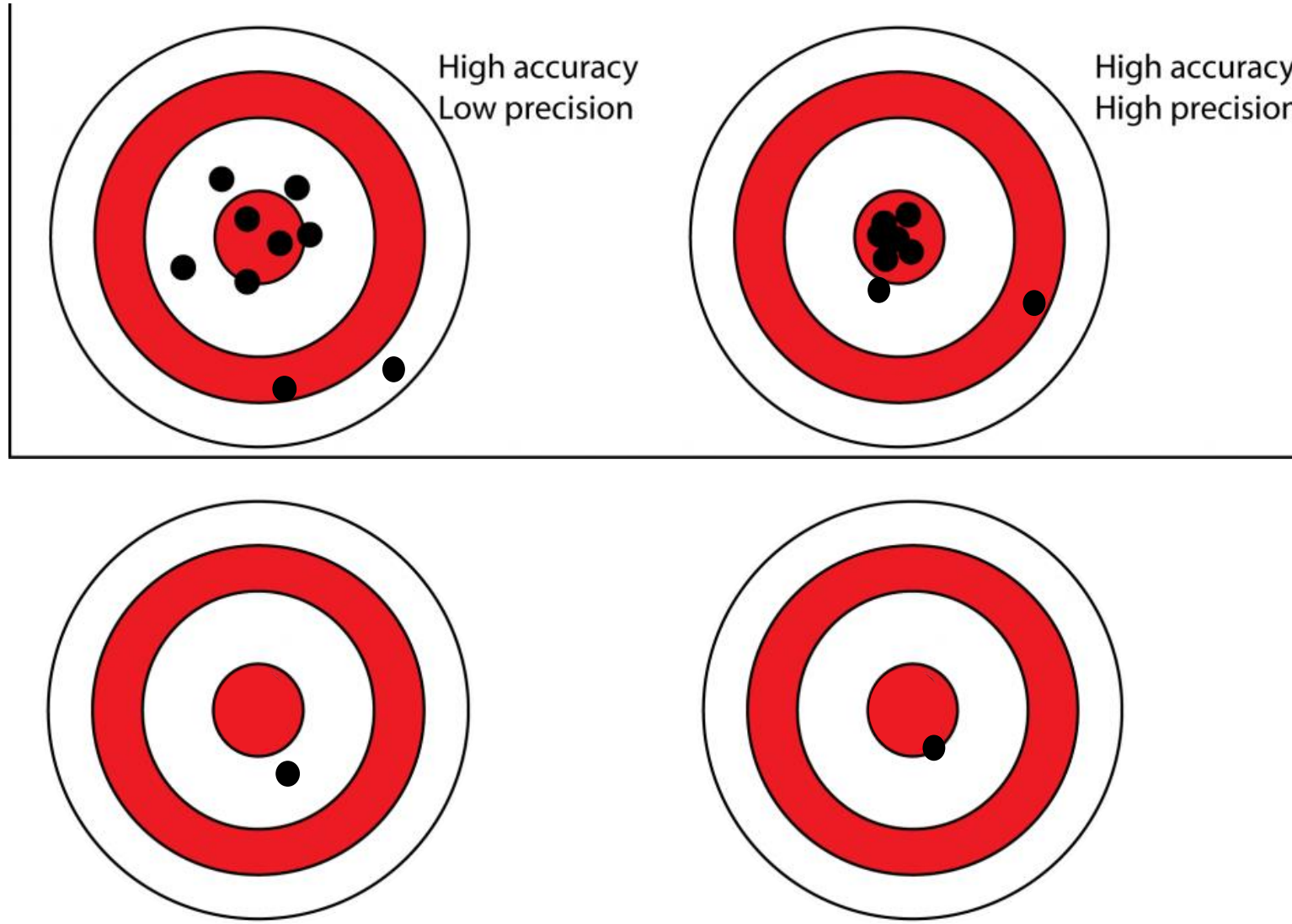
The expected value can be estimated by taking the mean of a set of data points.



$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i = \frac{x_1 + x_2 + \cdots + x_n}{n}$$

See `numpy.mean` (python)

Estimating Expected Values and Error

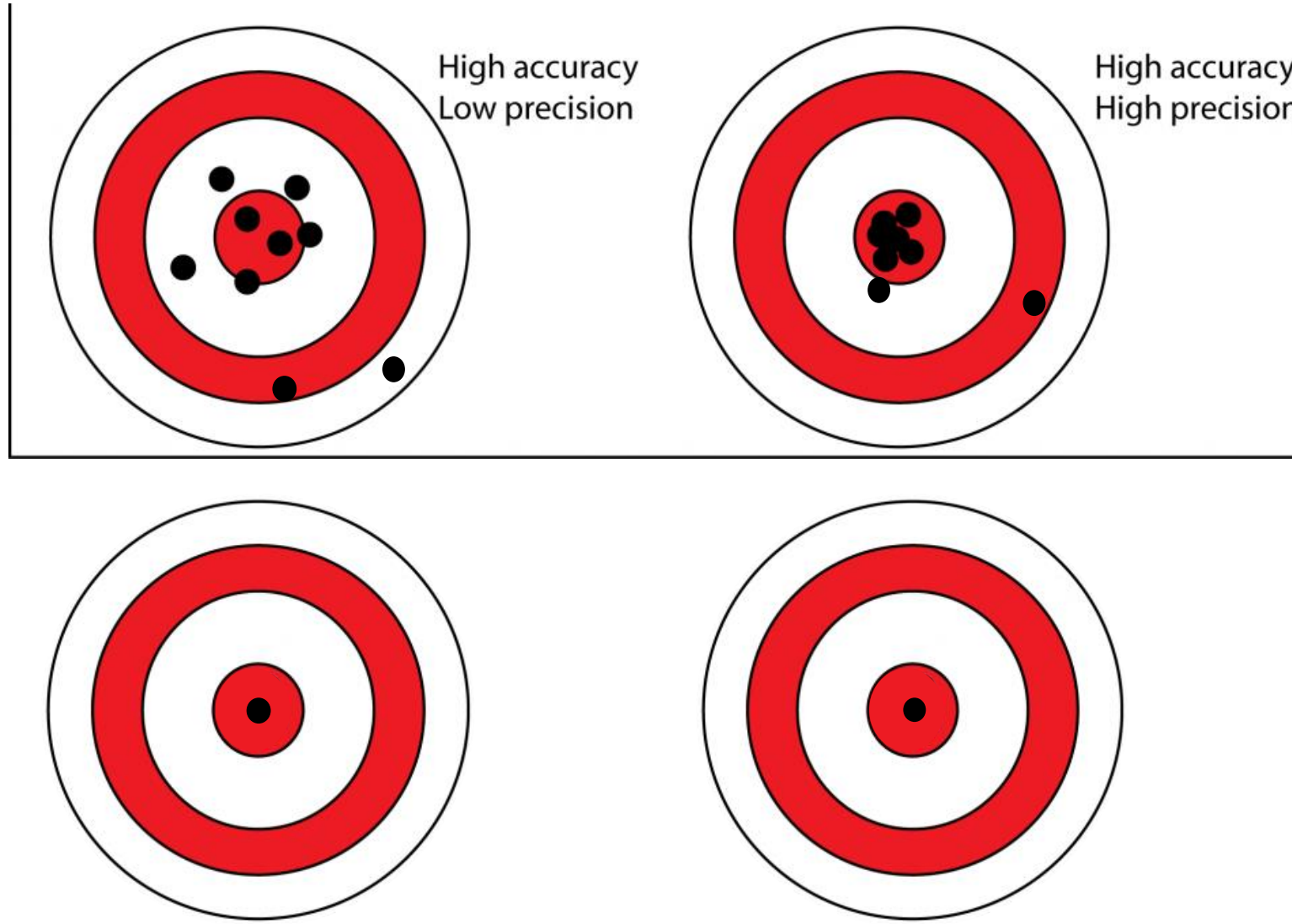


The median is sometimes more robust than the mean if there are many outliers.

Taking a **weighted average** is also an option to overcome outliers.

See `numpy.median` (python)

Estimating Expected Values and Error

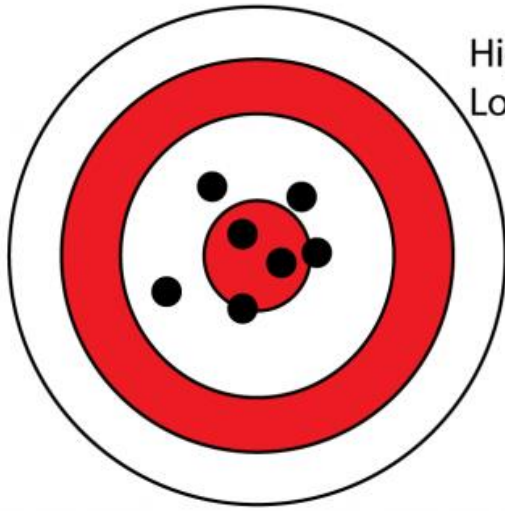


The median is sometimes more robust than the mean if there are many outliers.

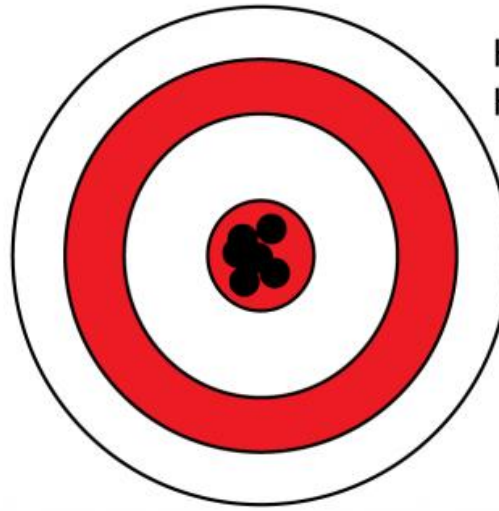
Taking a **weighted average** is also an option to overcome outliers.

See `numpy.median` (python)

Estimating Expected Values and Error

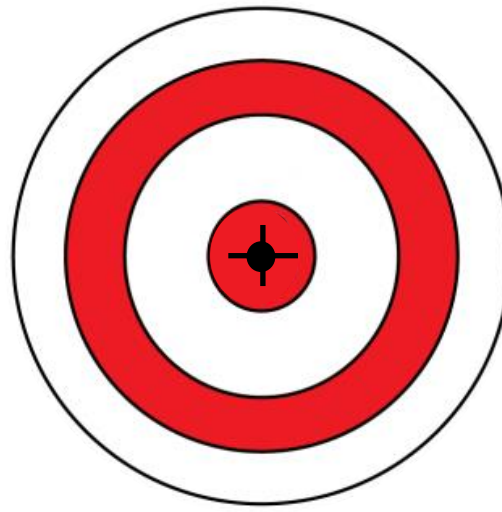


High accuracy
Low precision



High accuracy
High precision

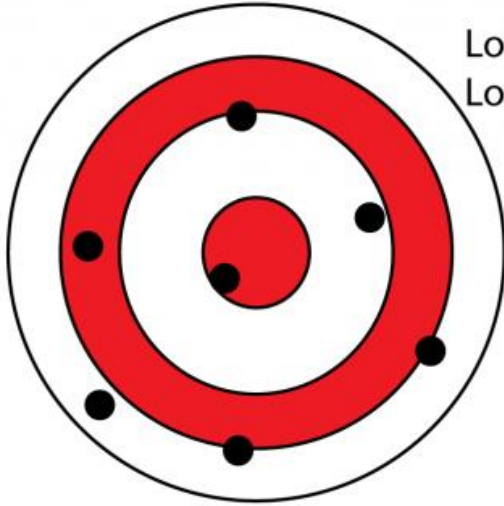
Precision or **error** can be estimated by taking the standard deviation of a set of data points.



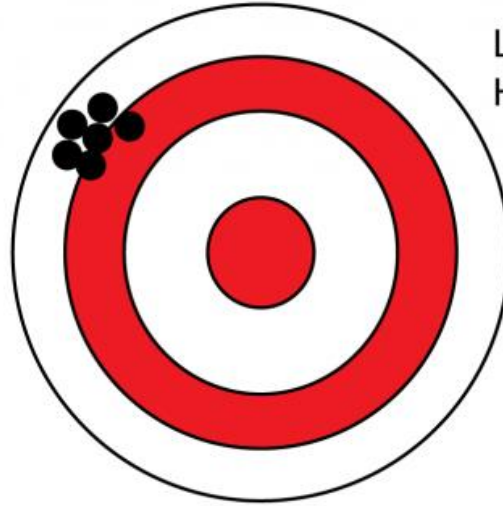
$$\sigma = \sqrt{\frac{1}{N} \sum_{i=1}^N (x_i - \mu)^2}$$

See `numpy.std` (python)

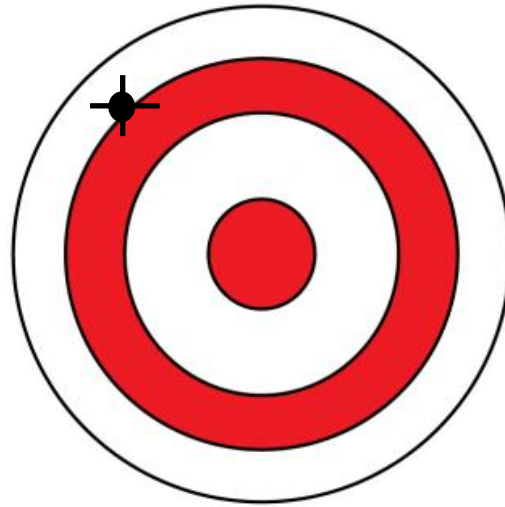
Estimating Expected Values and Error



Low accuracy
Low precision



Low accuracy
High precision

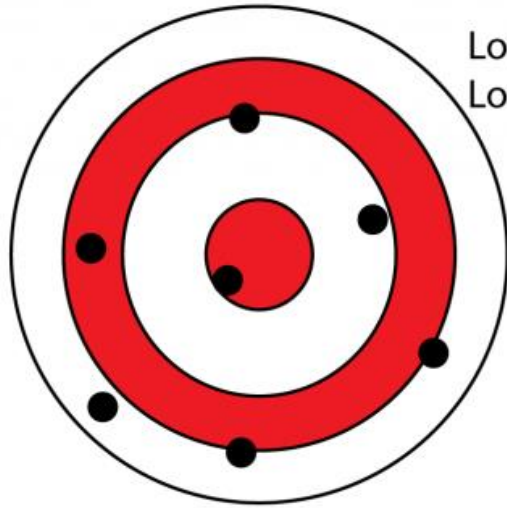


Precision or **error** can be estimated by taking the standard deviation of a set of data points.

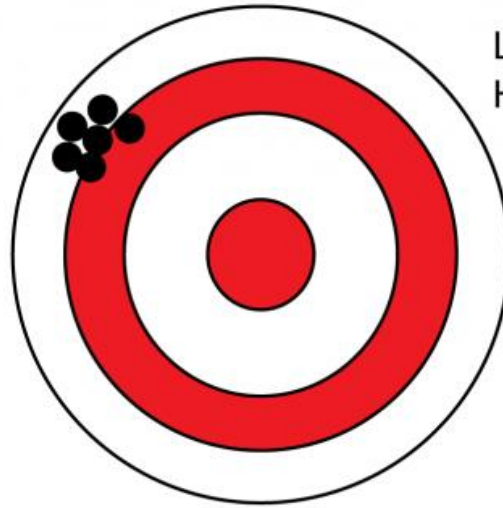
$$\sigma = \sqrt{\frac{1}{N} \sum_{i=1}^N (x_i - \mu)^2}$$

See numpy.std (python)

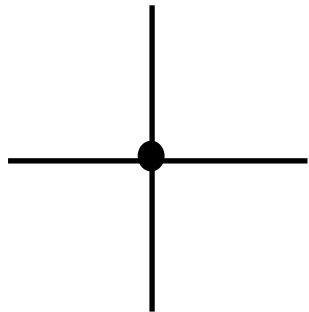
Estimating Expected Values and Error: Reality



Low accuracy
Low precision



Low accuracy
High precision

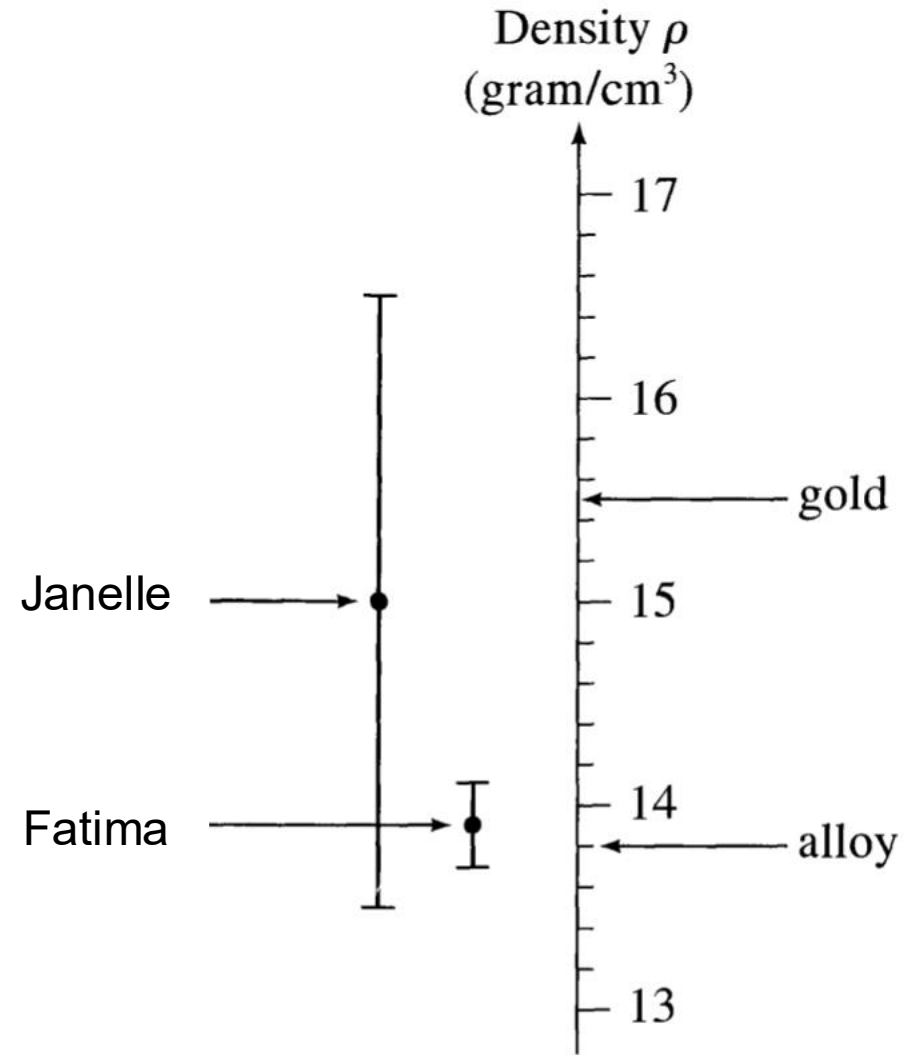


Precision or **error** can be estimated by taking the standard deviation of a set of data points.

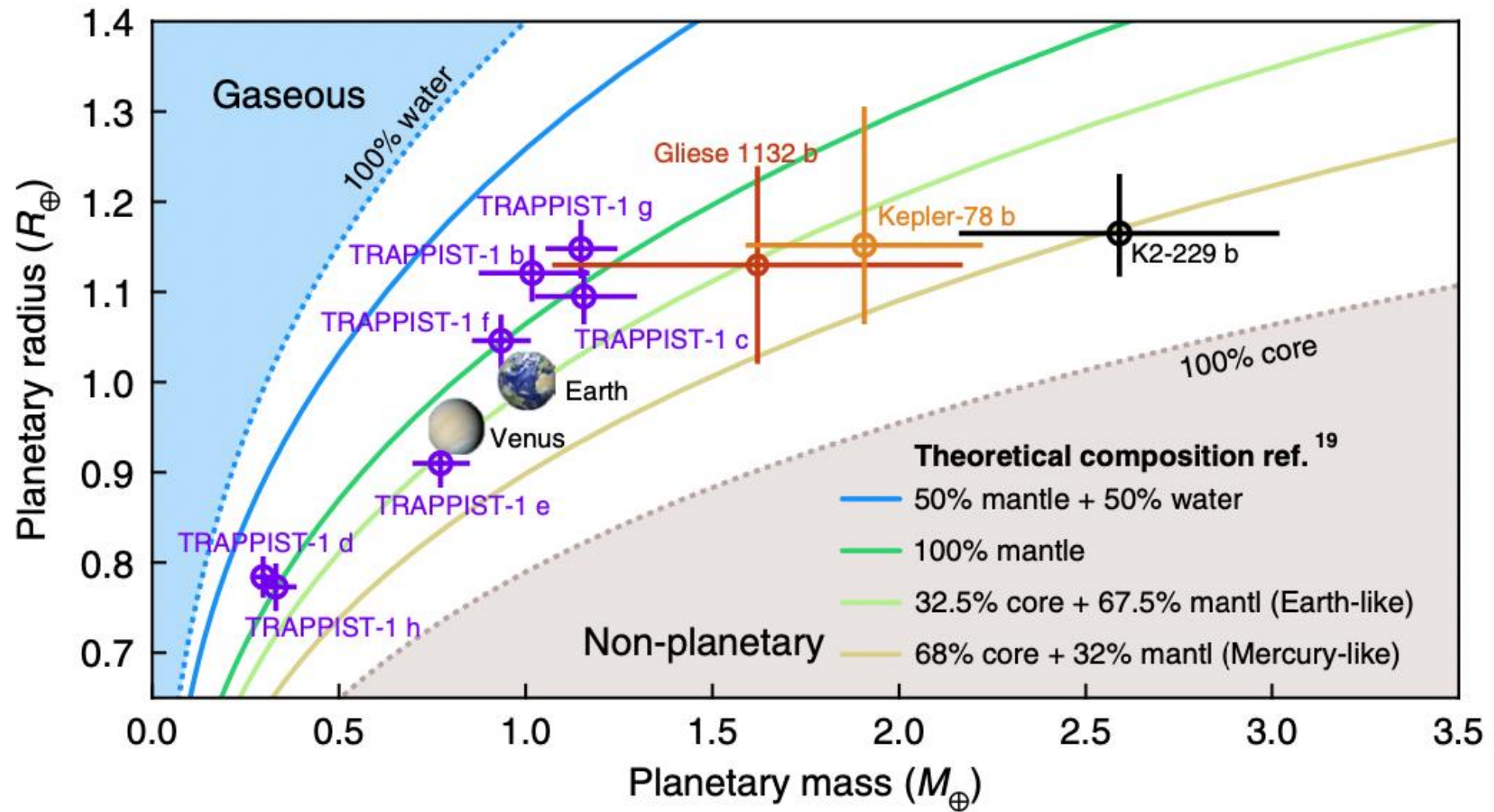
In observations we don't always know the ground truth so calibrations are important.

See `numpy.std` (python)

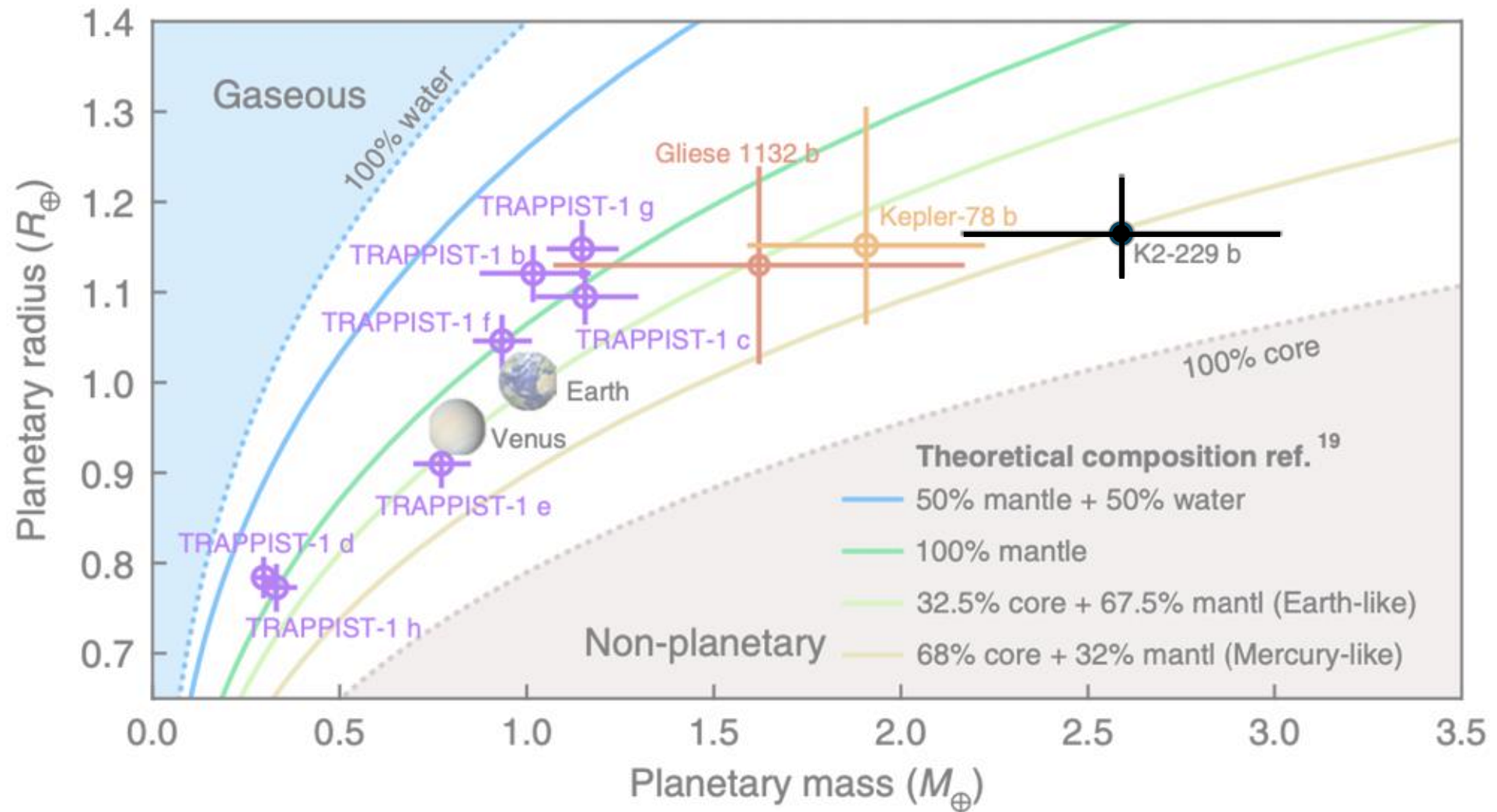
Interpreting Measurements with Error



Interpreting Measurements with Error



Interpreting Measurements with Error



Histograms

Histograms are graphs that communicate the occurrence rates or probabilities of observed values.

Histograms

Histograms are graphs that communicate the occurrence rates or probabilities of observed values.

Length Measurements (cm): 26, 24, 26, 28, 23, 24, 25, 24, 26, 25.

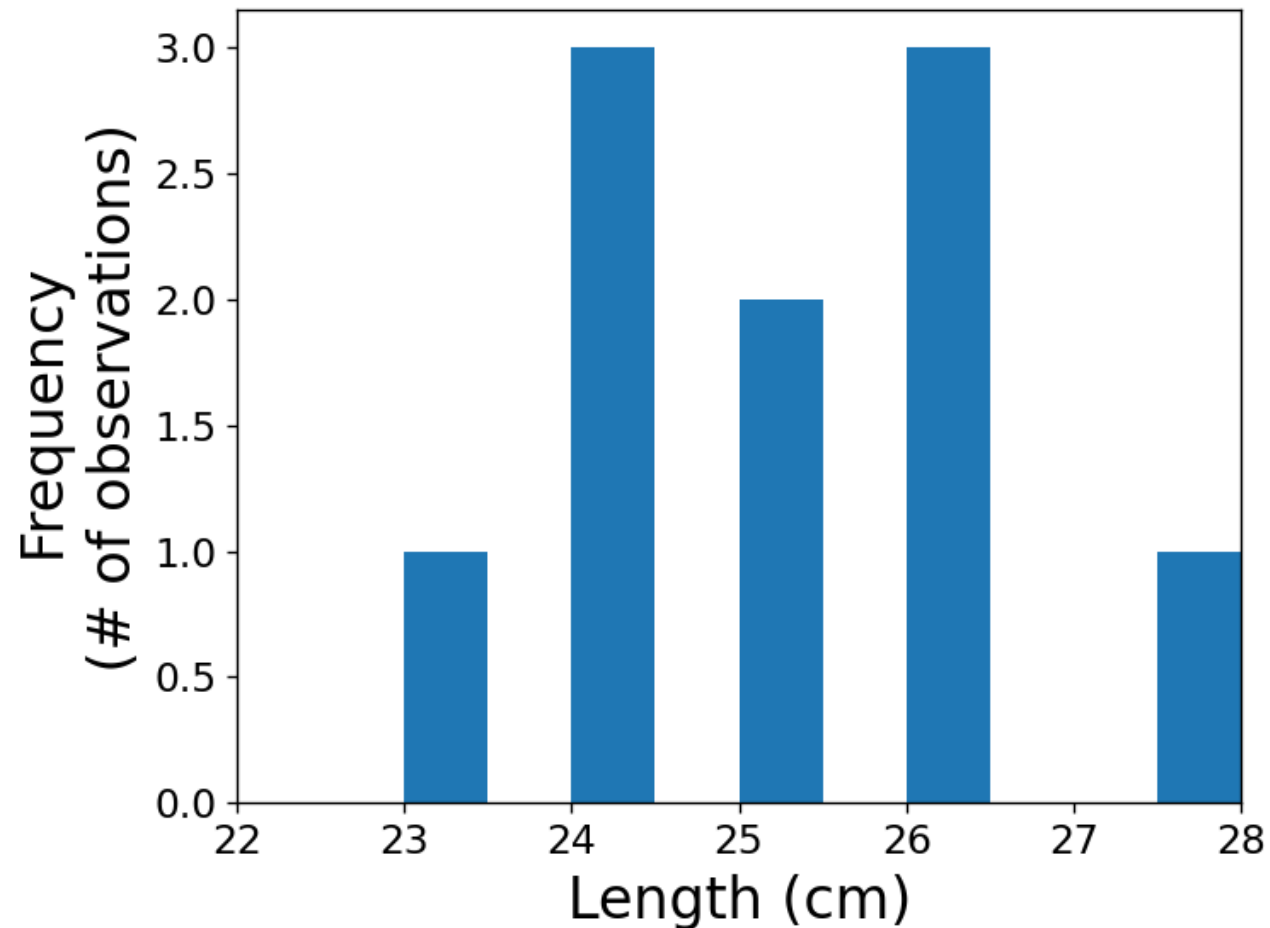
N Measurements: 10

Table 5.1. Measured lengths x and their numbers of occurrences.

Different values, x_k	23	24	25	26	27	28
Number of times found, n_k	1	3	2	3	0	1

Table 5.1. Measured lengths x and their numbers of occurrences.

Different values, x_k	23	24	25	26	27	28
Number of times found, n_k	1	3	2	3	0	1



`np.array(), plt.hist()`

Histograms

Histograms are graphs that communicate the occurrence rates or **probabilities** of observed values.

N Measurements: 10

$P(\text{bin}) = \text{Times Found} / N$

Table 5.1. Measured lengths x and their numbers of occurrences.

Different values, x_k	23	24	25	26	27	28
Number of times found, n_k	1	3	2	3	0	1
Probability	0.1	0.3	0.2	0.3	0.0.	0.1

Table 5.1. Measured lengths x and their numbers of occurrences.

Different values, x_k	23	24	25	26	27	28
Number of times found, n_k	1	3	2	3	0	1
Probability	0.1	0.3	0.2	0.3	0.0	0.1

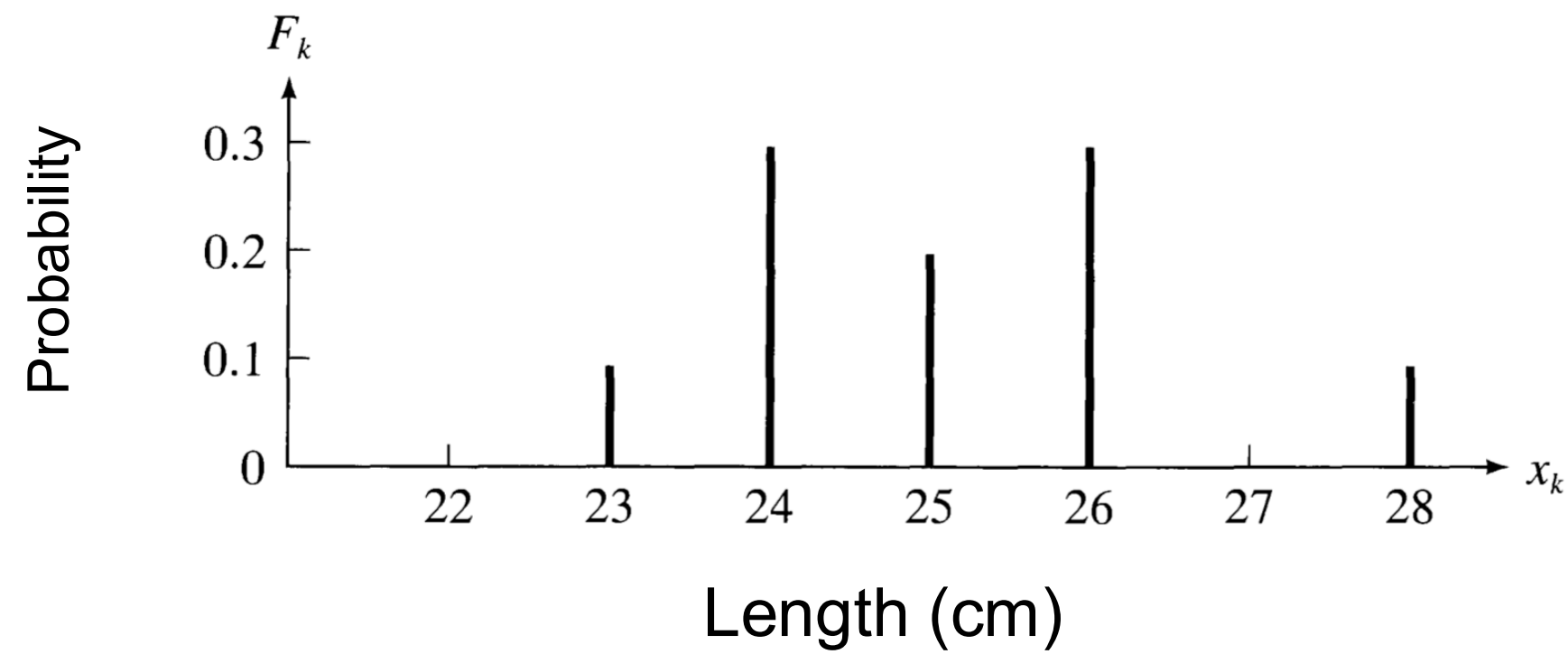
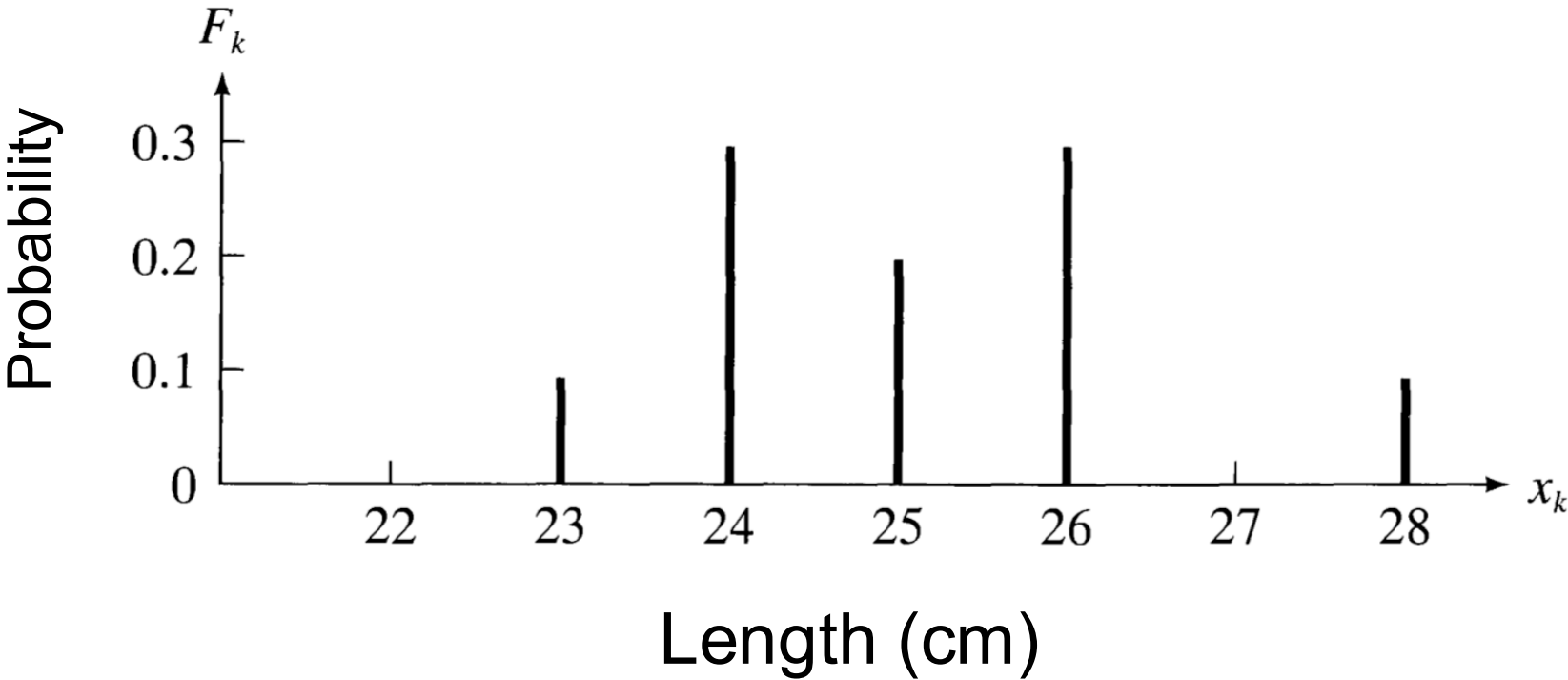


Table 5.1. Measured lengths x and their numbers of occurrences.

Different values, x_k	23	24	25	26	27	28
Number of times found, n_k	1	3	2	3	0	1
Probability	0.1	0.3	0.2	0.3	0.0	0.1

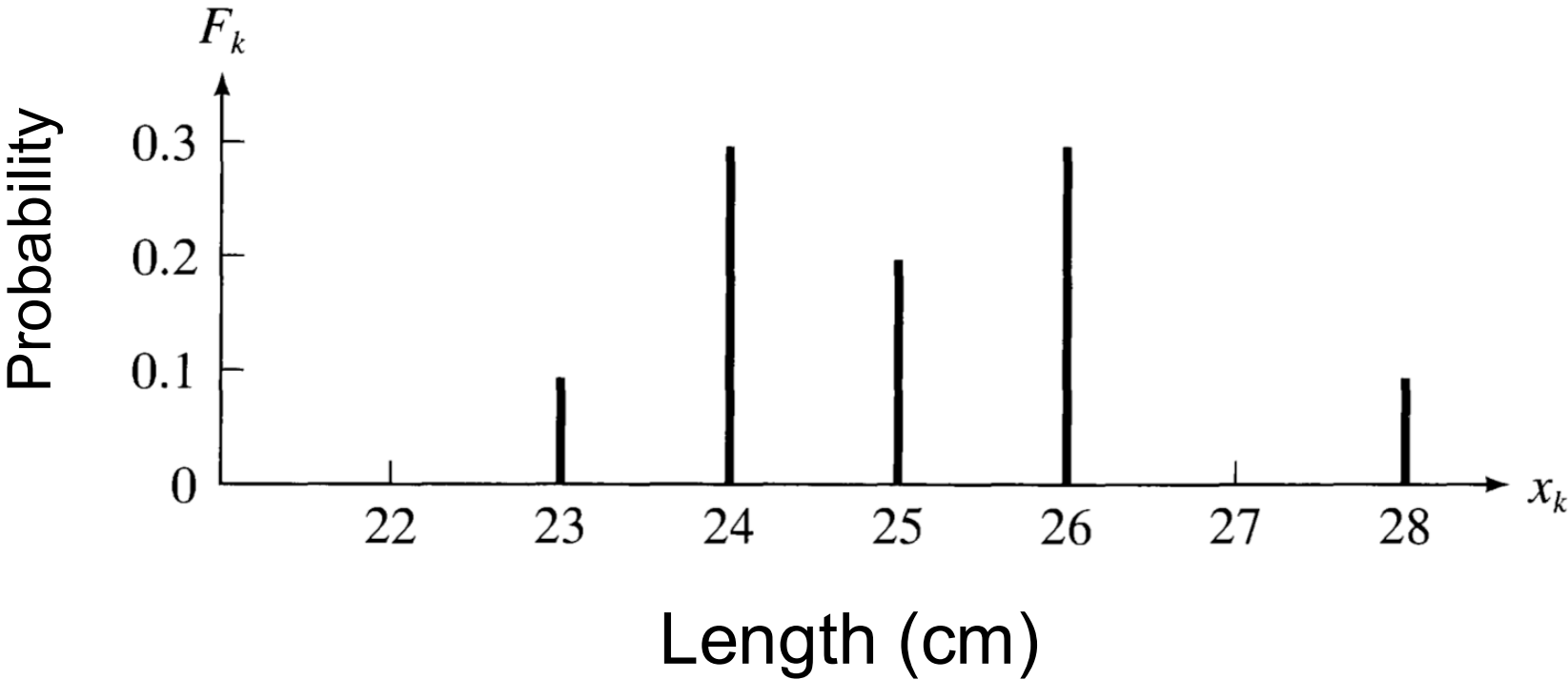


Mean =

Standard Deviation
 $\sigma =$

Table 5.1. Measured lengths x and their numbers of occurrences.

Different values, x_k	23	24	25	26	27	28
Number of times found, n_k	1	3	2	3	0	1
Probability	0.1	0.3	0.2	0.3	0.0	0.1

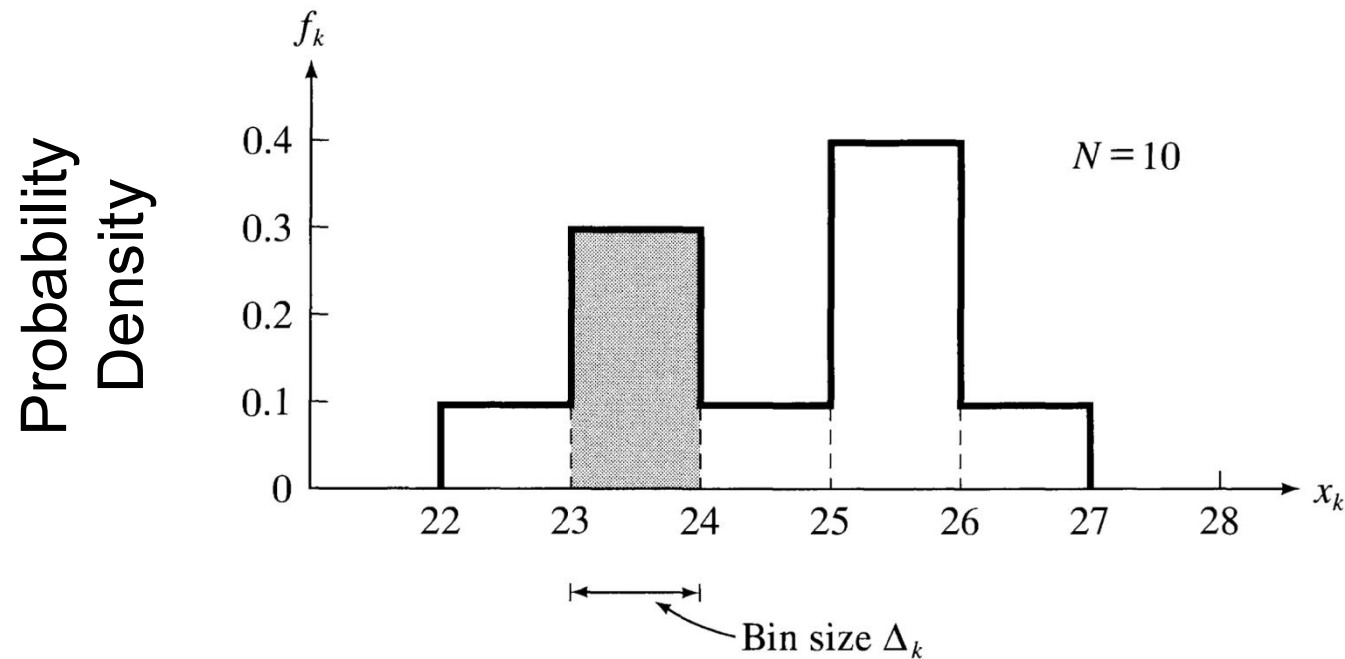


Mean = 25.1

Standard Deviation
 $\sigma = 1.37$

Histograms

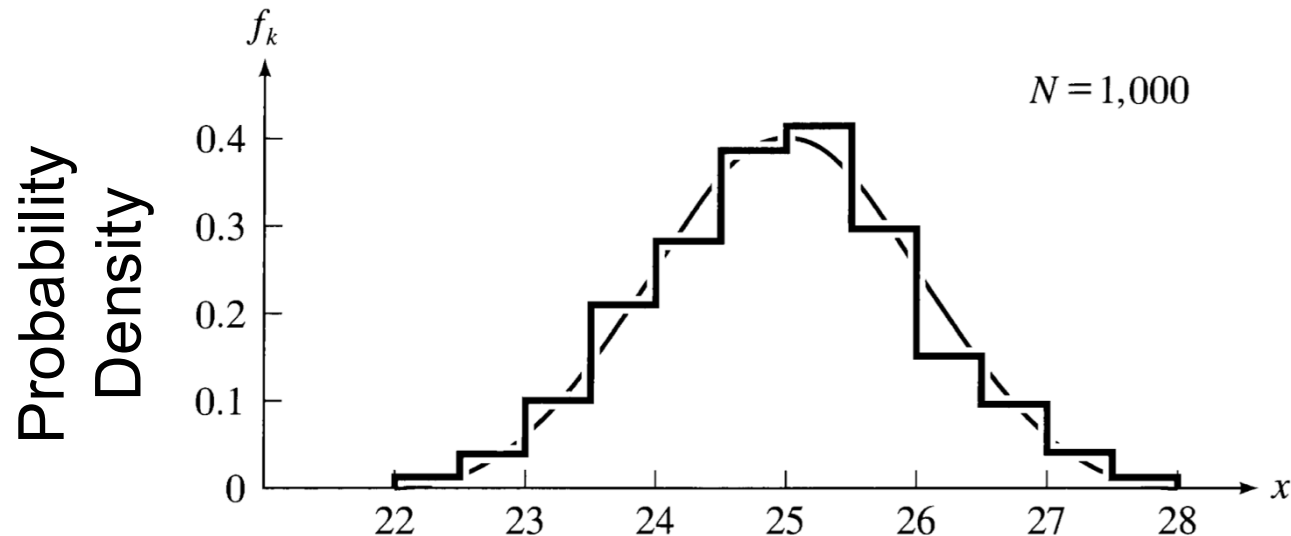
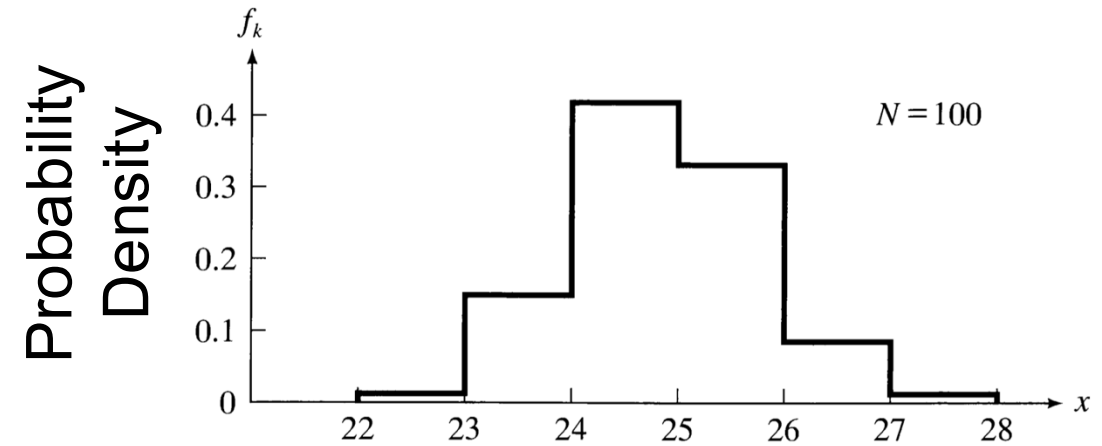
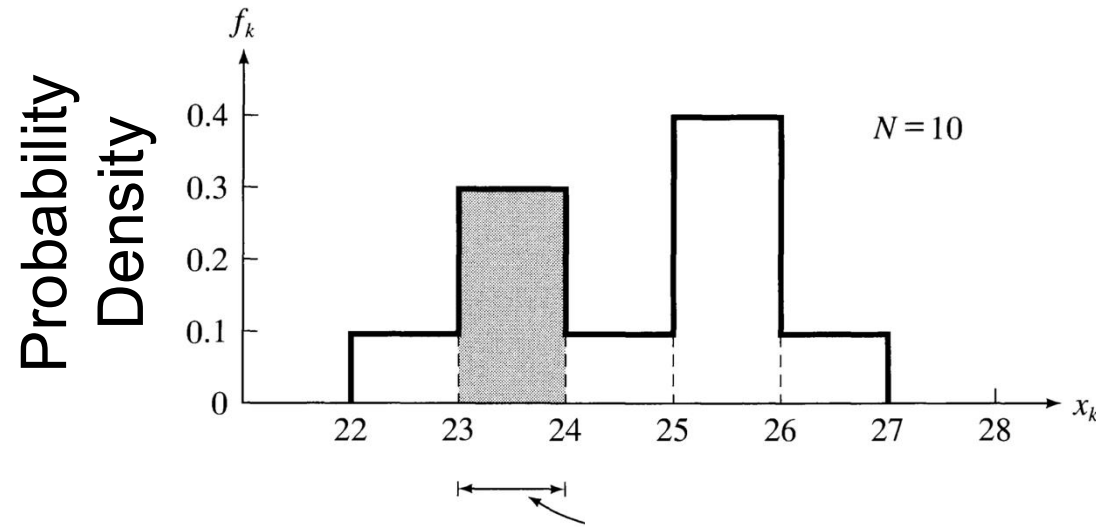
Histograms are graphs that communicate the occurrence rates or **probabilities** of observed values.



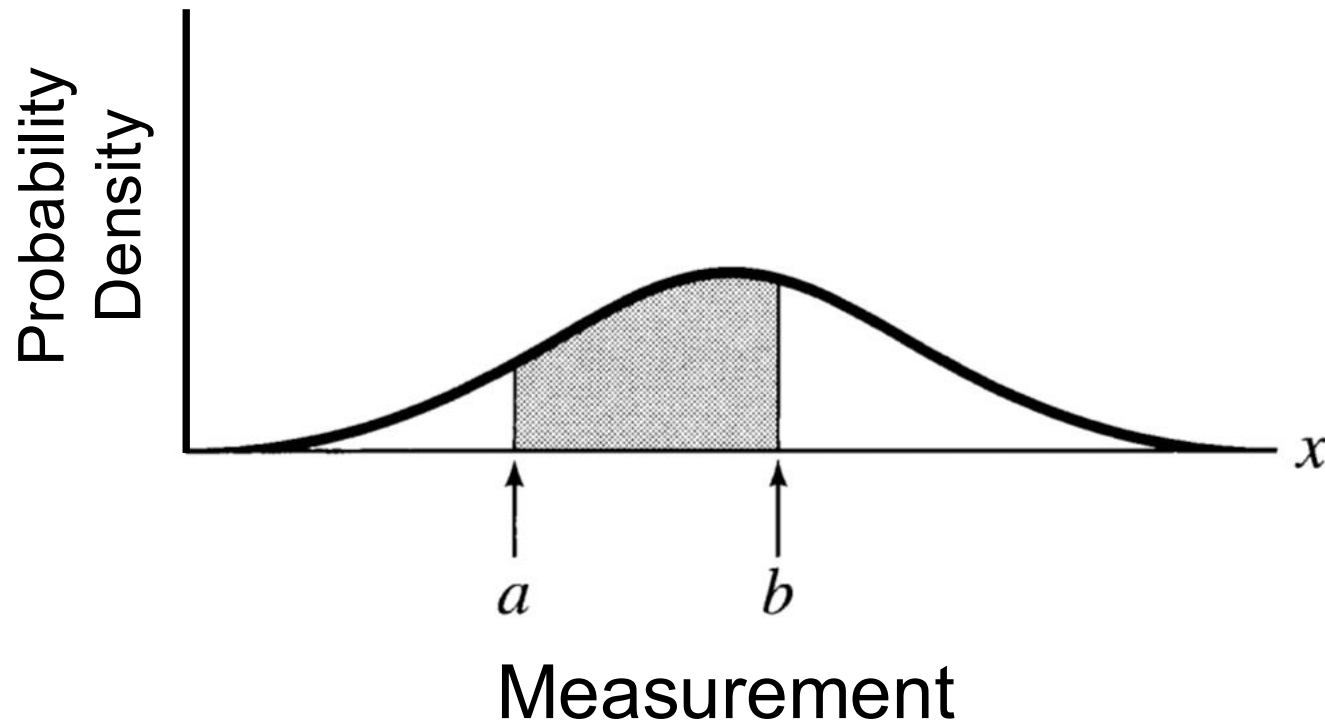
In bin histograms, the **probability density** is plotted vs. value ranges.

Limiting Distributions: Histograms where $N \rightarrow \infty$

For a large measurement number, a histogram appears smoother



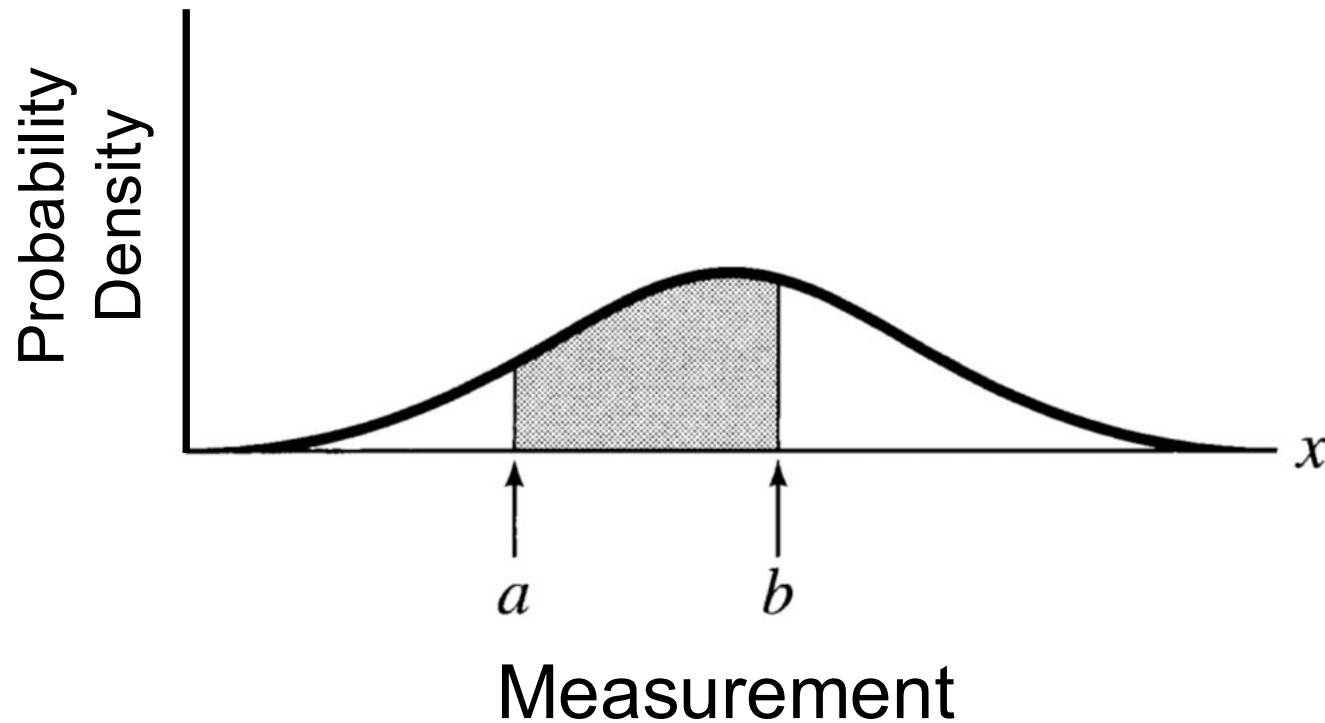
Calculating Probabilities of from Limiting Distributions



The area under a probability density curve between **a** and **b** gives you the probability of getting a measurement between those values.

$$\int_a^b f(x) \, dx$$

Calculating Probabilities of from Limiting Distributions

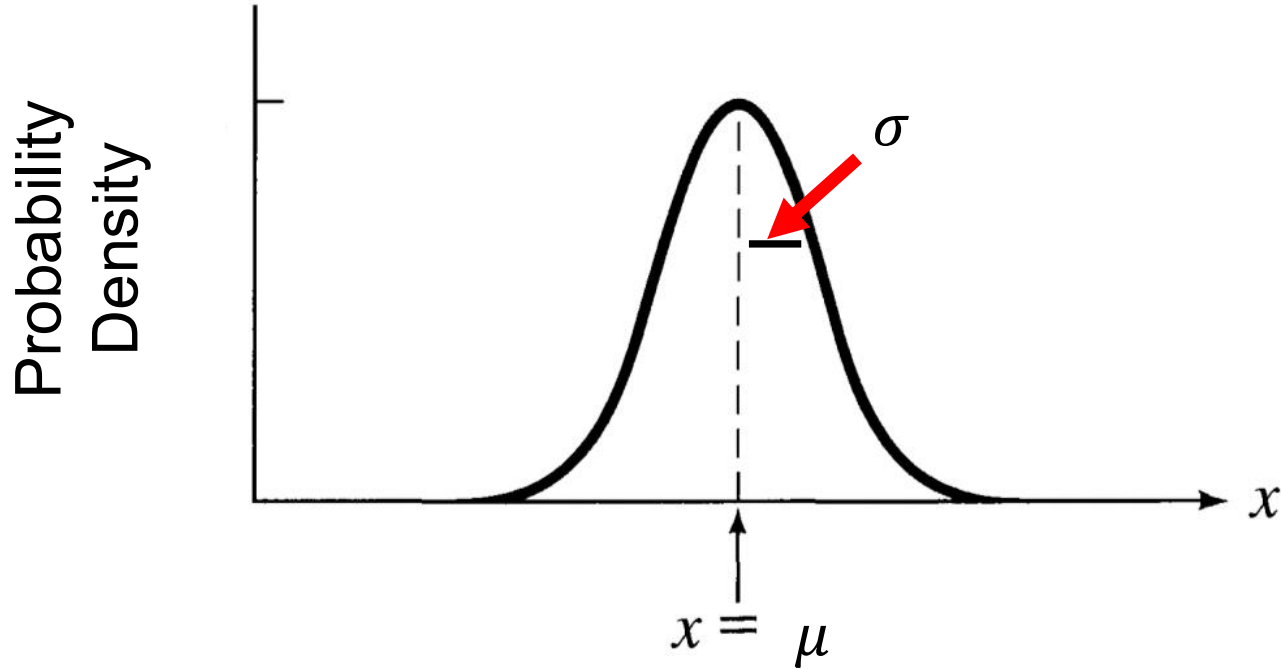


By extension, integrating over the entire probability density curve should give you a probability of unity.

$$\int_{-\infty}^{\infty} f(x) dx = 1.$$

If this is not true you need to normalize the probability density curve.

The Normal Distribution

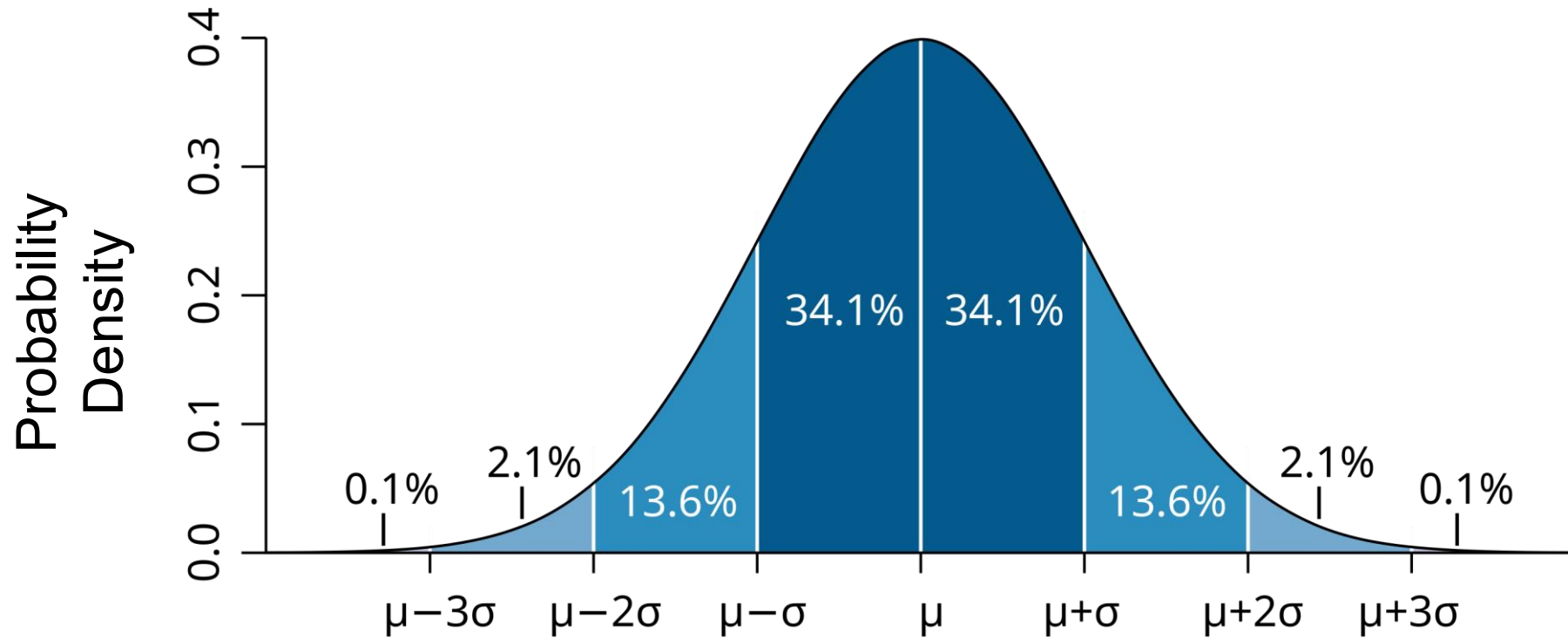


The Normal distribution or the Gaussian distribution is a bell-shaped probability curve centered on the value μ with a width parameter σ

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

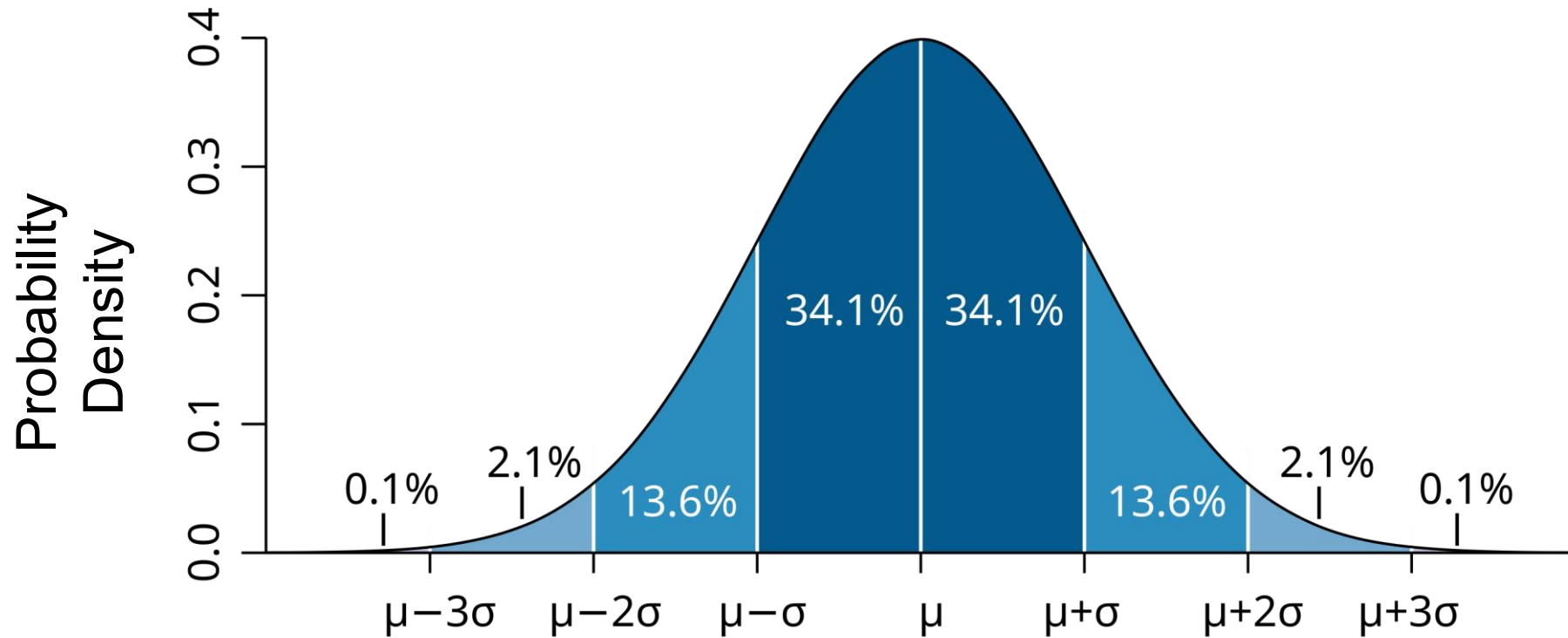
The coefficient in front satisfies the requirement that the area under the entire distribution must equal 1.

The Normal Distribution: Confidence Intervals



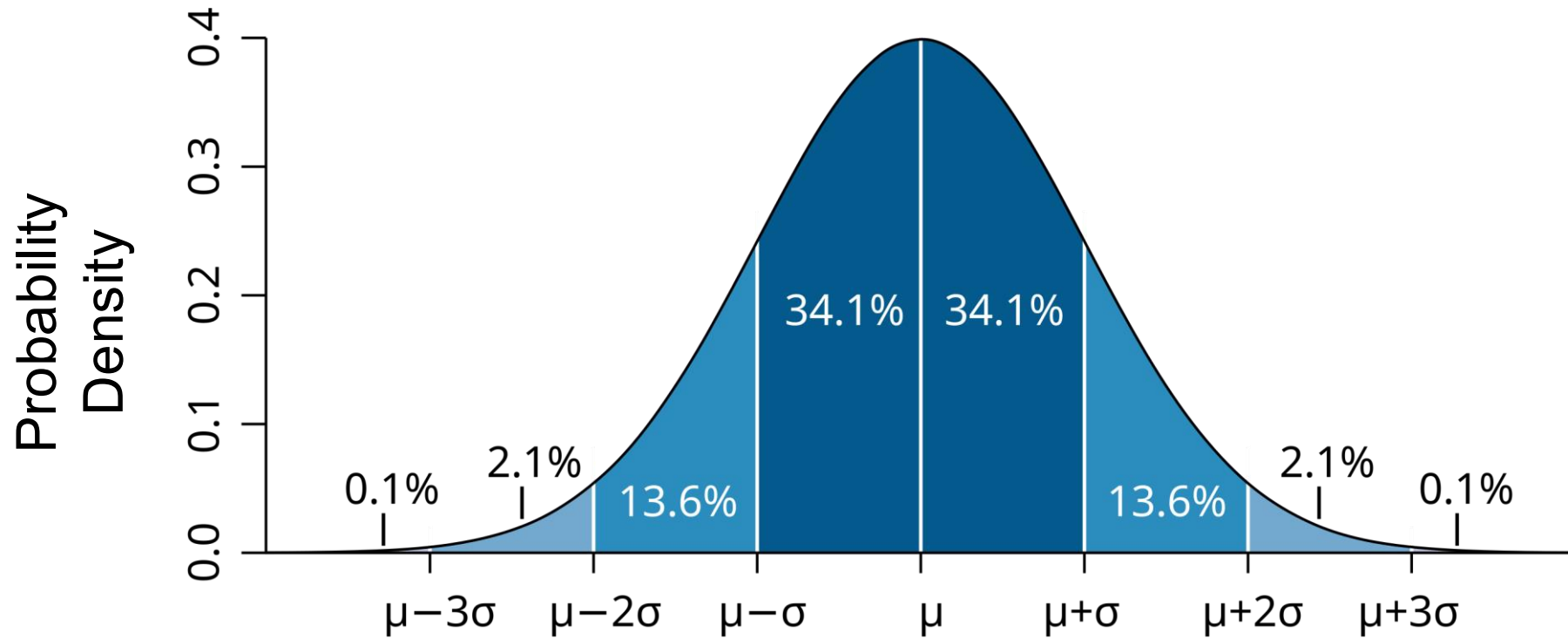
Confidence Interval is a range of values you need to observe (repeatedly) to estimate the true mean or standard deviation.

The Normal Distribution: Confidence Intervals



Between $\mu - \sigma$ and $\mu + \sigma$, 68.2% of the data is contained within this interval

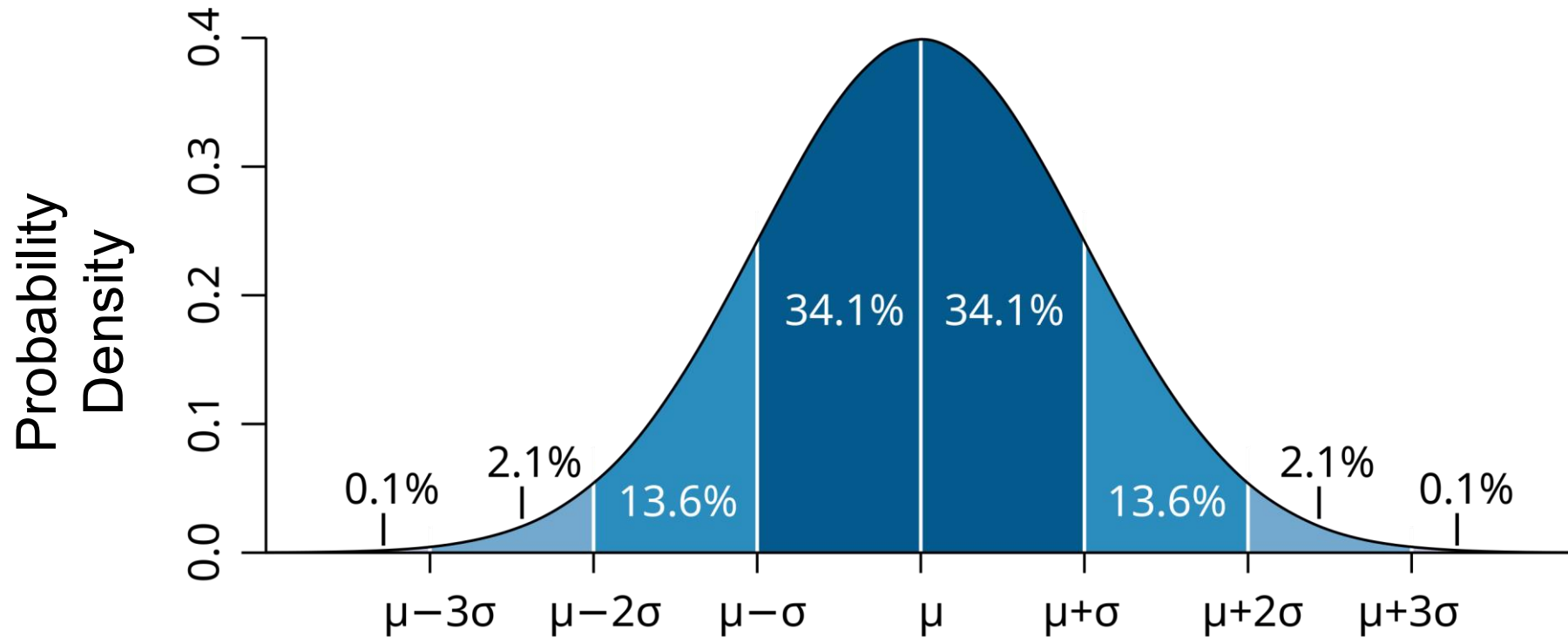
The Normal Distribution: Confidence Intervals



Between $\mu - \sigma$ and $\mu + \sigma$, 68.2% of the data is contained within this interval

Between $\mu - 2\sigma$ and $\mu + 2\sigma$, 95.4% of the data is contained within this interval

The Normal Distribution: Confidence Intervals



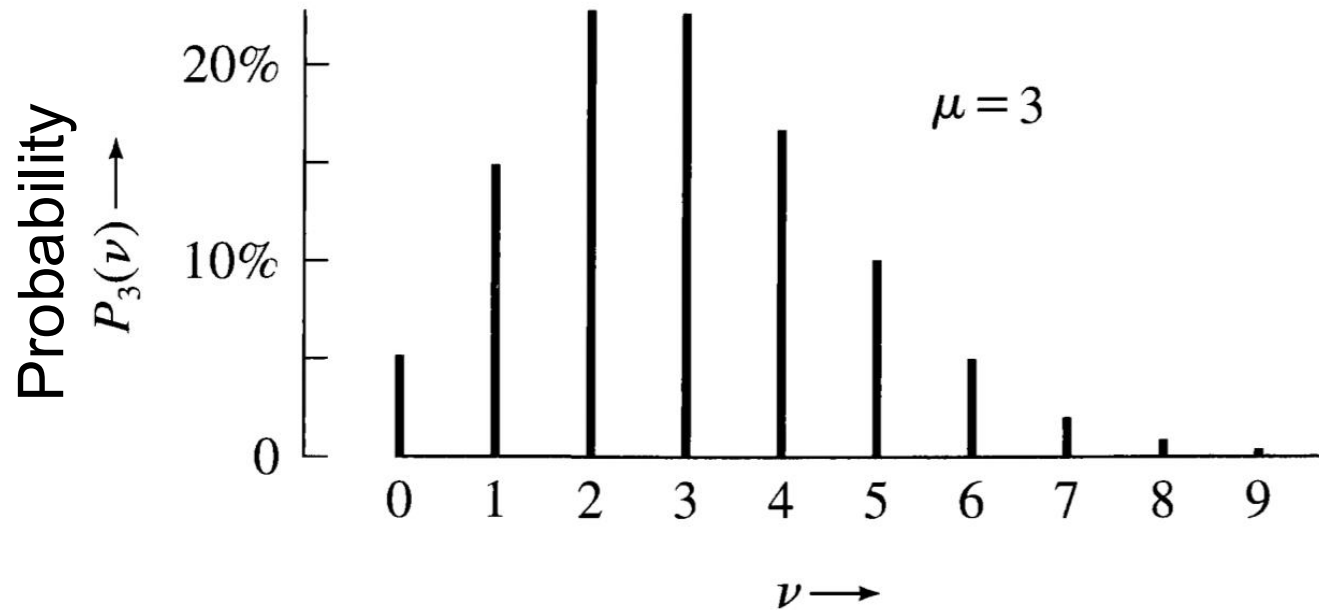
Between $\mu - \sigma$ and $\mu + \sigma$, 68.2% of the data is contained within this interval

Between $\mu - 2\sigma$ and $\mu + 2\sigma$, 95.4% of the data is contained within this interval

Between $\mu - 3\sigma$ and $\mu + 3\sigma$, 99.7% of the data is contained within this interval

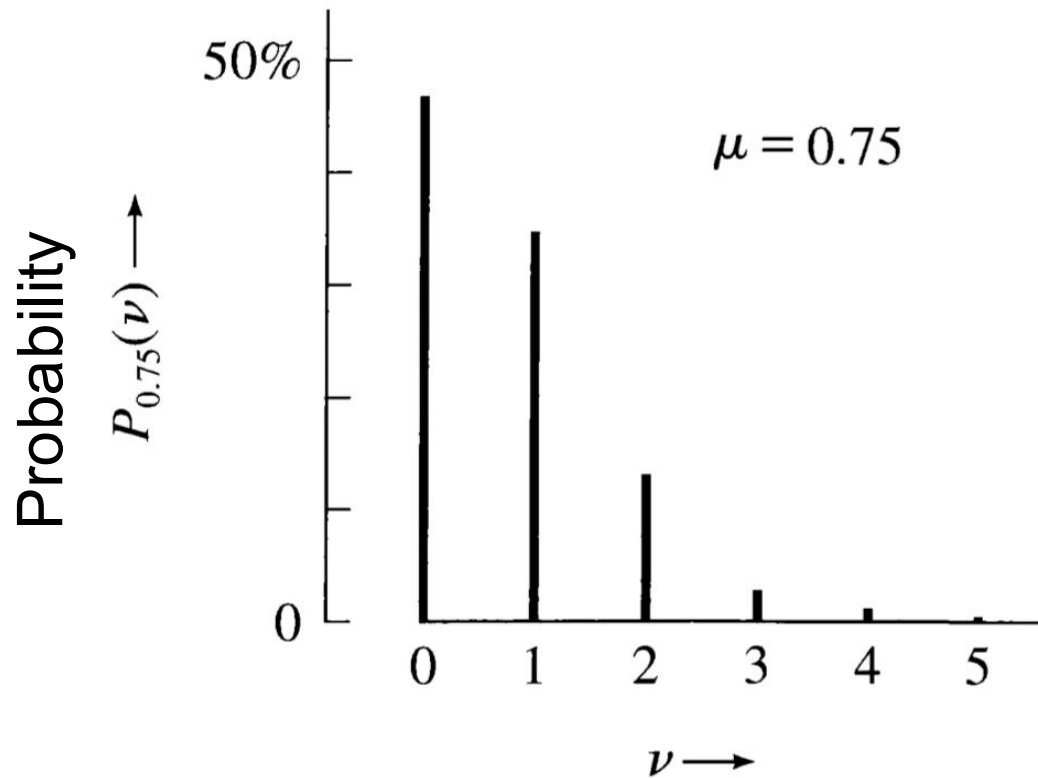
The Poisson Distribution

The Poisson distribution describes the probability of observing N (ν) discrete events over a fixed interval of time assuming some expected average (μ).



$$P_{\mu}(\nu) = e^{-\mu} \frac{\mu^{\nu}}{\nu!}.$$

The Poisson Distribution

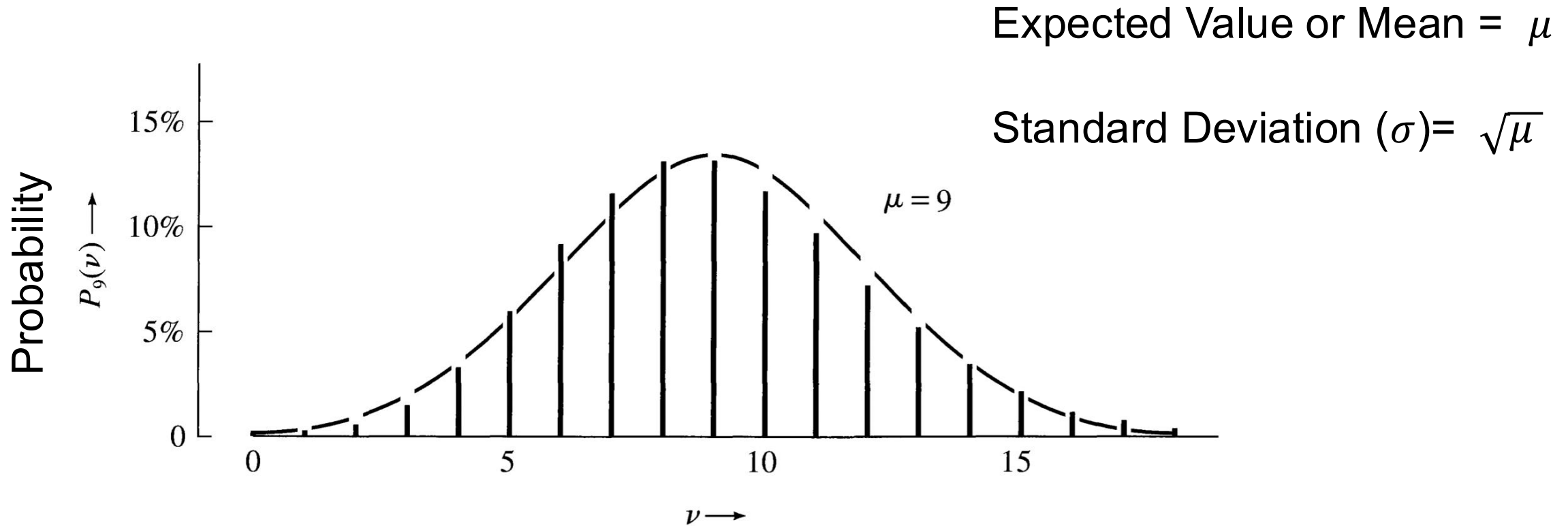


The Poisson distribution describes the probability of observing $N(\nu)$ discrete events over a fixed interval of time assuming some expected average (μ).

$$P_{\mu}(\nu) = e^{-\mu} \frac{\mu^{\nu}}{\nu!}.$$

Unlike the Normal/Gaussian distribution the expected values never go below zero. You either count something or you don't.

The Poisson Distribution in the Large N Limit

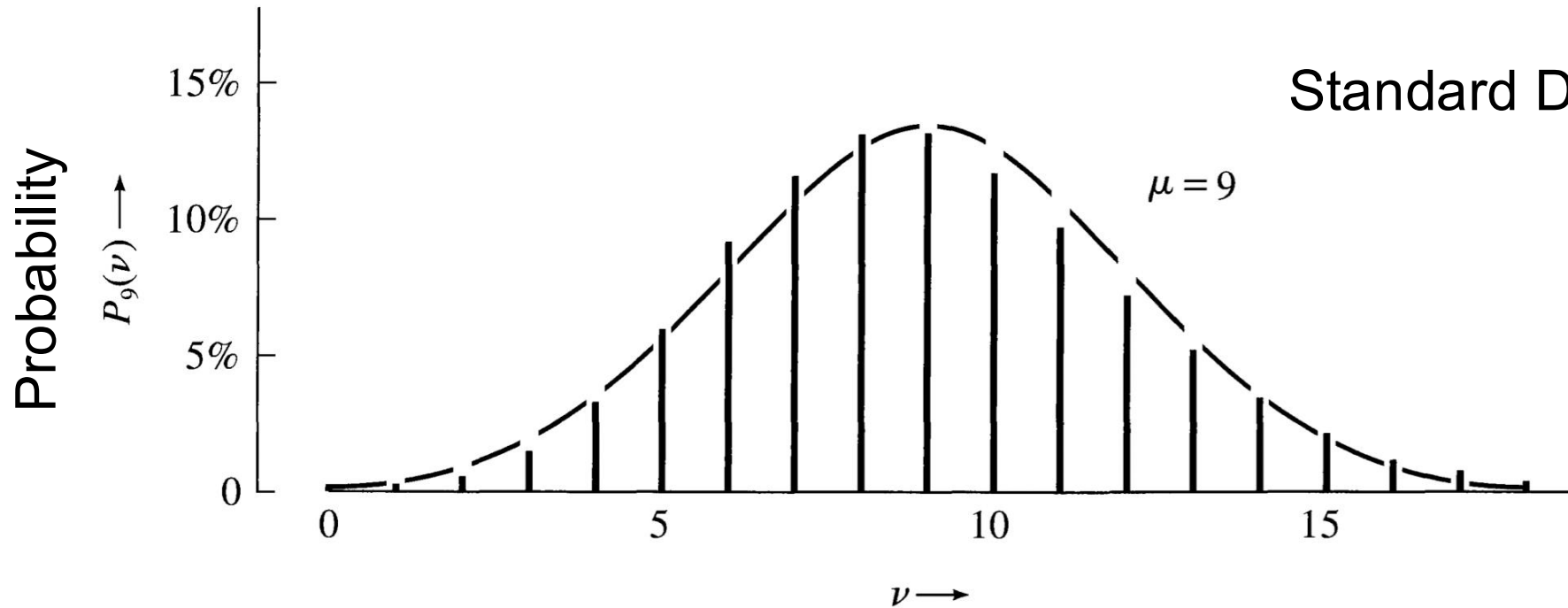


For a sufficiently high event rate, the Poisson distribution behaves like a Gaussian distribution where both the peak and width are set by the expected count number.

The Poisson Distribution in the Large N Limit

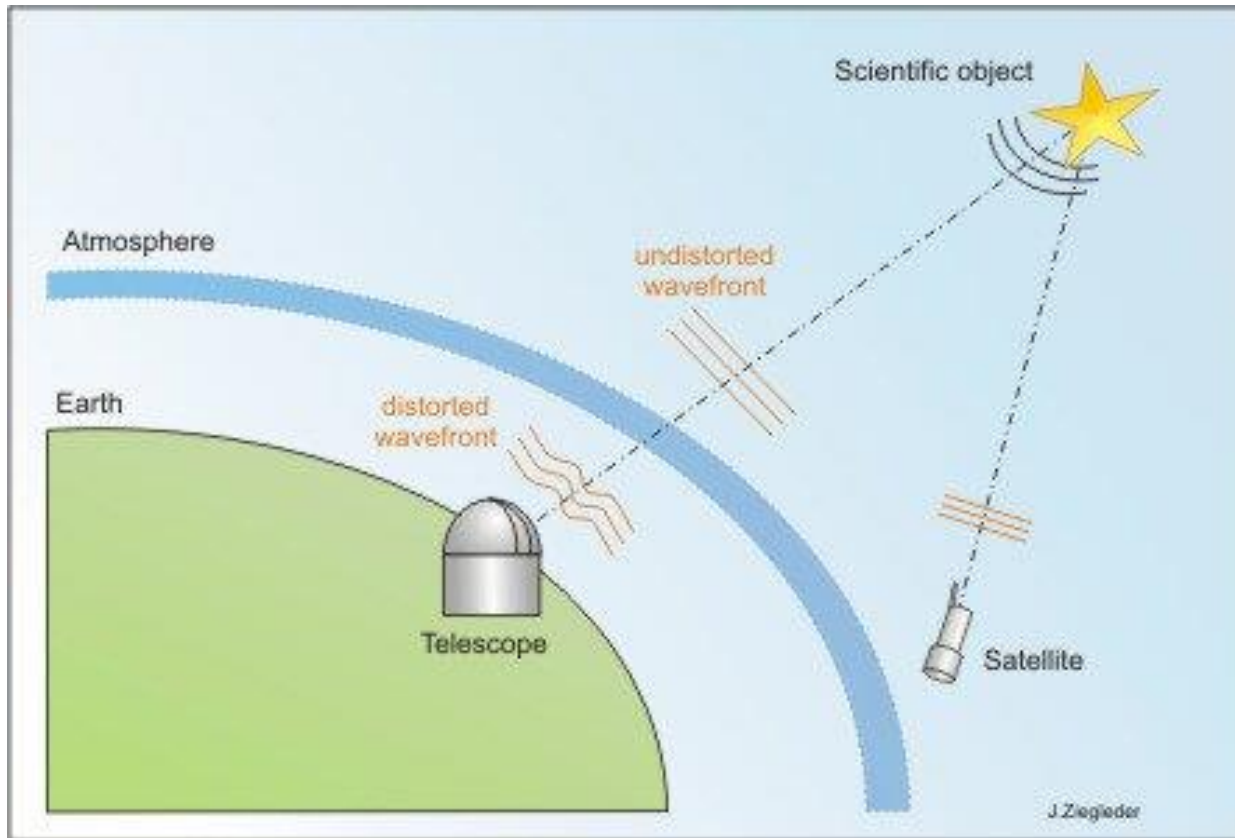
Expected Value or Mean = μ

Standard Deviation (σ) = $\sqrt{\mu}$



Friday in class we will explore the Poisson, Normal and other Distributions via python.

Sources of Error in Astronomical Observations

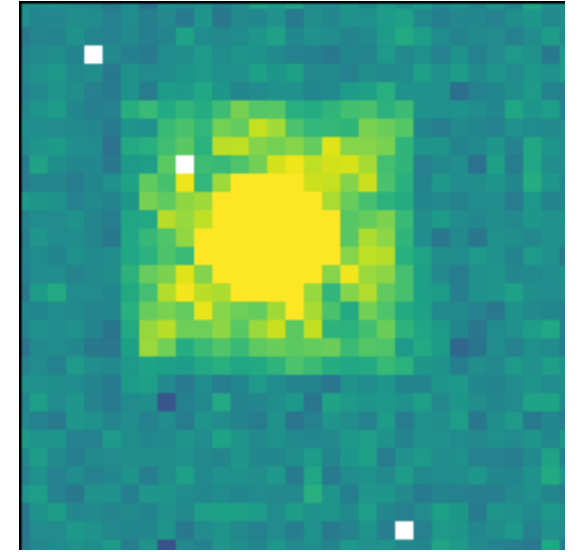


1. Random Noise from the Source
2. Earth's Atmosphere (Ground-based)
 - Sky/Airmass
 - Water Vapor (Infrared)
3. Optics/Materials from the telescope
4. Noise from your sensor/receiver/detector

Random Noise from Photons

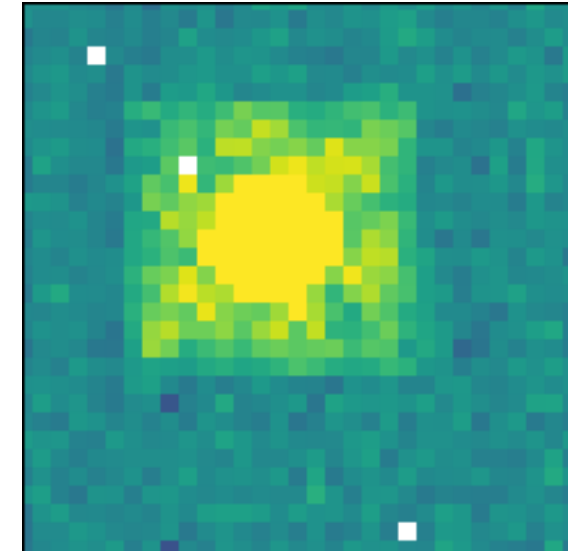
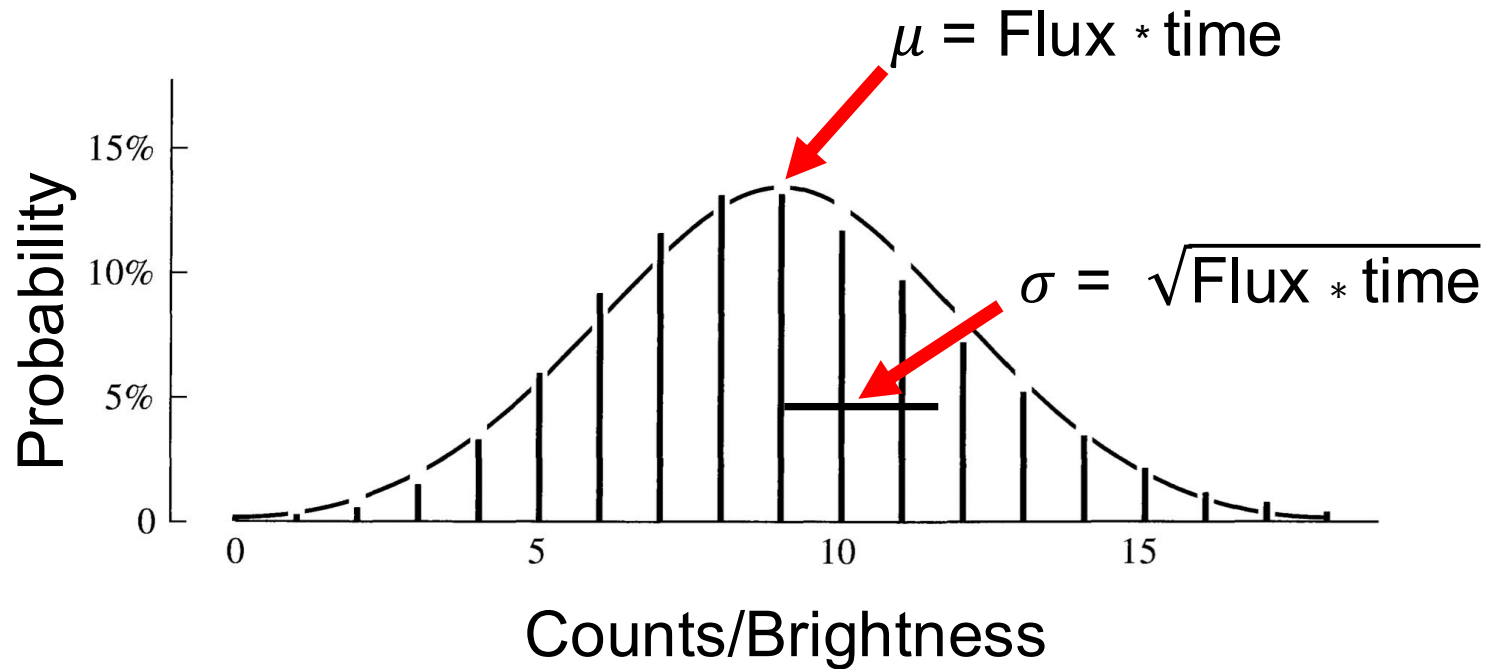
- Random noise from astronomical sources is often caused by the random behavior of **photons**.
- Photons are discrete particles that are emitted or reflected from a source. Either 0, 1, 2.. etc photons
- The Poisson distribution describes the probability of observing N events over a fixed interval of time assuming some fixed event rate.
- For a sufficiently high event rate, the Poisson distribution behaves like a Gaussian distribution.

Photon Noise Limit Application: Bright Objects



**Raw JWST Acquisition Image
From Bright Object Time Series
Mode**

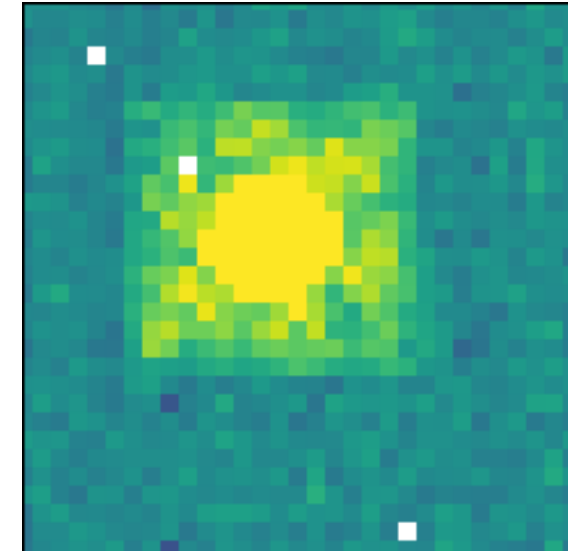
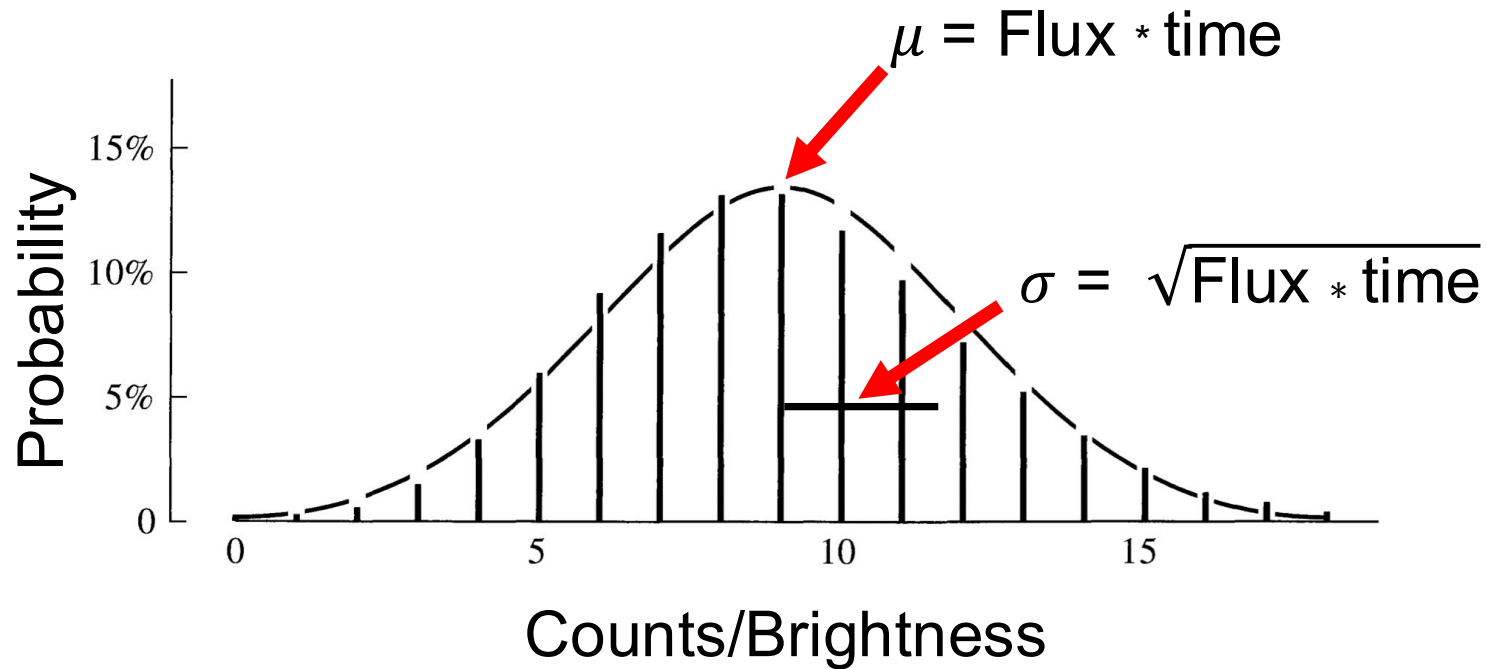
Photon Noise Limit Application: Bright Objects



Expected Value or Mean (μ) = Flux * time

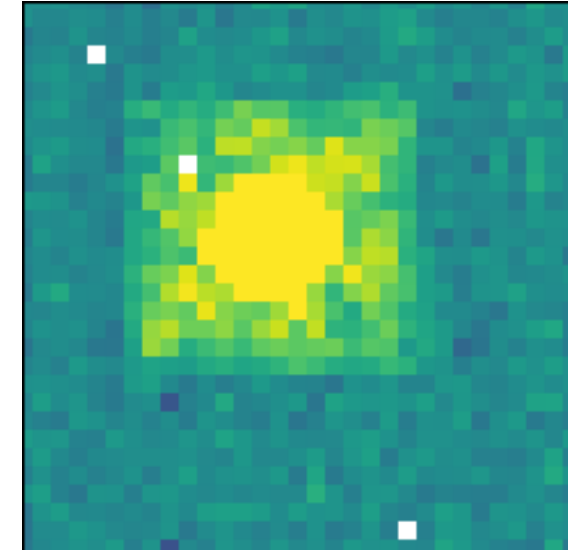
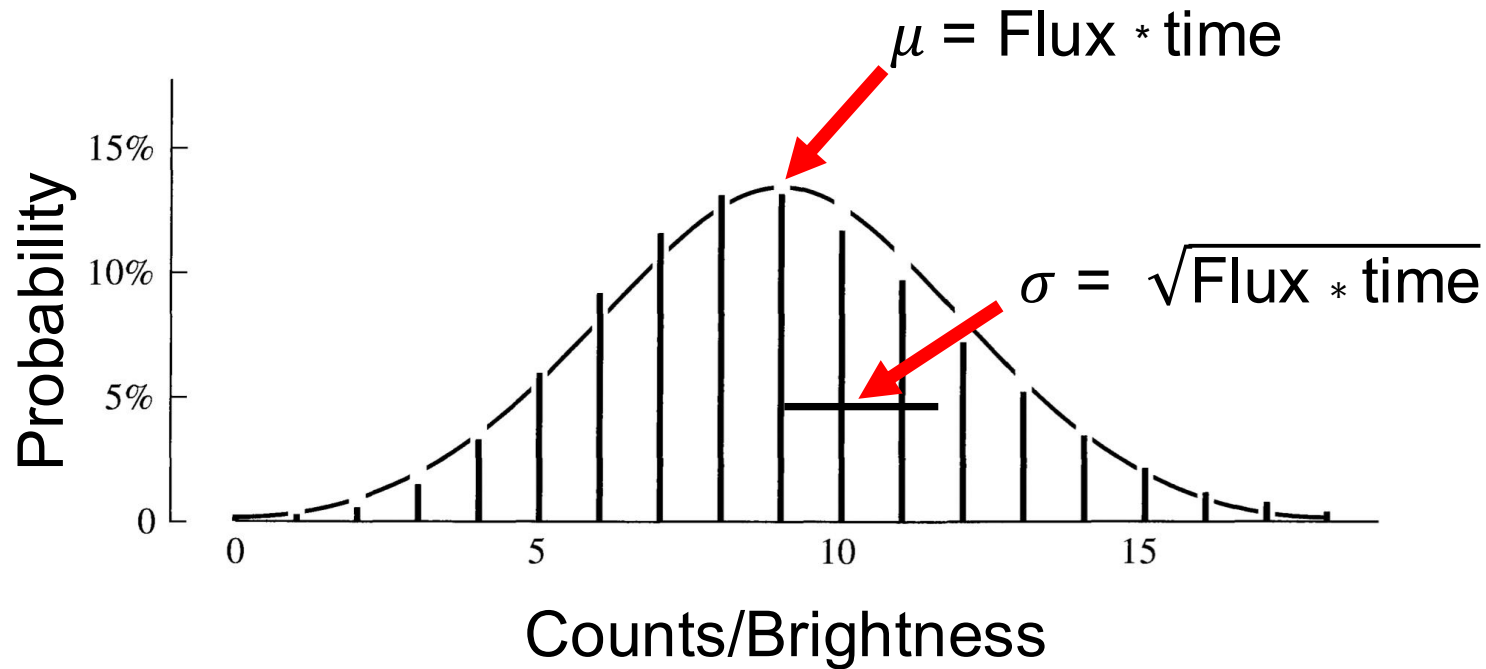
Standard Deviation or Error (σ) = $\sqrt{\text{Flux} * \text{time}}$

Photon Noise Limit Application: Bright Objects



If you observe for a longer time the mean value gets bigger, but the standard deviation gets bigger at a slower rate. More time leads to a more precise measurement.

Photon Noise Limit Application: Bright Objects



Telescope time is precious. Just because you can get an ultra precise measurement doesn't mean you should. You need to estimate how much time is necessary for the question you are posing.